

Rational Voter Learning, Issue Alignment, and Polarization*

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Abstract

We model electoral competition between two parties when voters can rationally learn about their political positions through flexible information acquisition. Rational voter learning generates polarized and aligned political preferences, even when voters' true positions are unimodally distributed and independent across policy issues. When parties strategically select their positions to influence voter learning, party and voter polarization increase as information costs decline, and parties may adopt positions more extreme than their ideal policies. These results arise from two new forces introduced by endogenous voter learning: parties gain from moderating to skew learning in their favor, but the more extreme party also gains from polarizing to trigger more voter learning.

Keywords: rational inattention, voter ideology, electoral competition, polarization

JEL Classifications: D72, D83, D43, L13

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1 Introduction

Voter positions in the United States display two puzzling features. First, one can predict a voter’s position on most policy issues remarkably well by knowing just their location on a one-dimensional left-right axis. Voter positions on different issues are strongly aligned, which is surprising given the wide variety of issues, such as taxation, immigration, and the environment. Second, evidence suggests that voter positions are increasingly polarized, in the sense of being clustered around two poles on the left-right axis.¹

This paper provides a joint explanation of issue alignment and polarization based on rational voter learning. Previous research and public discourse attribute issue alignment and polarization to voter biases like confirmation bias, herding due to echo chambers, or partisan news media. By contrast, we show issue alignment and polarized ideology emerge naturally from rational (i.e., no biases) and individual (i.e., no herding) voter learning, driven by voters’ own information choices (i.e., no media effects). Central to the mechanism is that voters learn about their political position through flexible information acquisition in order to decide between two parties. Voter learning can involve understanding the effects of policies, for example, learning about the effects of tariffs to inform their position on trade policy. We assume that such learning is costly, whether in time, effort, or money. This paper shows that the resulting cost-minimization motive structures rational voter learning in a way that generates issue alignment and polarization.

Our model based on rational learning can help disentangle behaviors driven by biases from those arising from rational ideology formation. This distinction is important for assessing the functioning of democratic elections (Achen and Bartels 2017). If biases dominate, candidates may cater to these biases rather than advancing policies that address societal needs. Conversely, if voter positions reflect rational learning, we may be more optimistic that elections produce policies aligned with voters’ interests. Whether this optimism is warranted is explored in the second part of this paper, which studies the effects of voter learning on party positions.

We analyze rational voter learning in an otherwise standard political-economy model. To study issue alignment, we incorporate a multidimensional policy space, where each dimension represents a different policy issue. Voters face two parties, each adopting a policy platform in this space. These policy platforms are exogenous in the first part of our paper. Following the literature, a voter’s utility decreases quadratically in the distance between a policy and her *ideal point*, which reflects her political position. In our model, voters initially lack knowledge of their ideal point but can form political opinions through learning. We assume that voters start with a homogeneous prior that conforms to the true distribution of ideal points, which in the simplest case is normal

¹Section 1.1 examines the evidence for issue alignment and ideological polarization in detail. While the evidence for issue alignment is strong, the evidence on bimodality is more speculative; there is ongoing debate about this issue (Fowler, Hill, Lewis, Tausanovitch, Vavreck, and Warshaw 2022). Understanding the sources of potential polarization is important because of evidence of increasing voter polarization (Pew Research Center 2014; Martin and Yurukoglu 2017; Eguia and Hu 2022) and because of the relevance of polarization for political tensions, as emphasized by the literature on polarization (Esteban and Ray 1994): a polarized distribution can lead to the formation of two homogeneous groups with few individuals bridging the divide, increasing the risk of conflict.

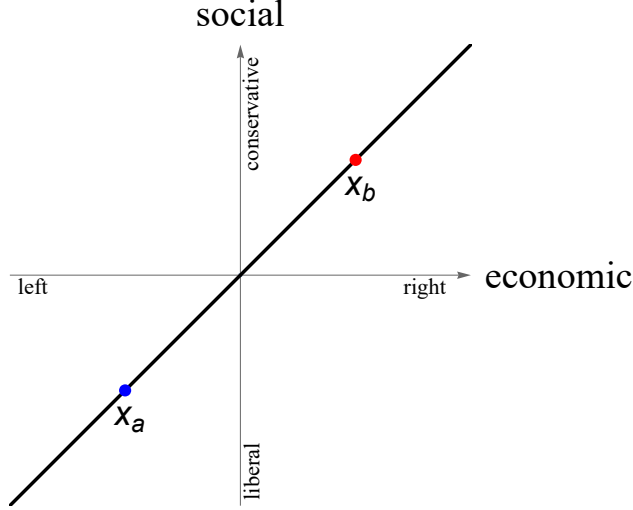


Figure 1: Example policy platforms x_a and x_b of parties a and b , respectively, under two policy issues. Voter learning induces revealed ideal points on the diagonal line.

and independent across issues.² Voters then update their beliefs by acquiring information flexibly, following the rational-inattention framework (Sims 2003). This framework allows voters to choose both how much and what type of information to acquire, subject to a cost increasing in information. After acquiring information, and given quadratic utility, a voter selects the party whose policy bundle is closer to her expected ideal point. We refer to this expected ideal point as her *revealed ideal point*, distinguishing revealed ideology—the distribution of revealed ideal points induced by learning—from the distribution of true ideal points.

Our first result is that rational voter learning generates issue alignment and polarization: revealed ideal points align along a left-right axis and cluster around two poles, even if true ideal points are independent across issues and centrally clustered. At the core of this mechanism is that voters trade off making an informed choice and minimizing the cost of information. By acquiring only the information necessary to determine which party position is closer, voters align their preferences along a single axis and polarize, a process we explain in more detail next.

To understand the high-level intuition for why cost-effective learning generates issue alignment, consider a two-dimensional policy space with an economic and a social issue. Suppose the two parties propose policy platforms, x_a and x_b , as shown in Figure 1: one platform is more economically left and socially liberal than the other. A unique line passes through these platforms, the direction of which we refer to as the *axis of disagreement*. This axis of disagreement is the direction along which parties differentiate, combining the two issues in proportion to the extent of disagreement on each. Crucially, voters learn only where their ideal point lies along this axis, as this determines which platform is closer. Their position along directions orthogonal to the axis is irrelevant, as it does not affect the *relative* proximity of the platforms. Consequently, voters align preferences across issues, learning only whether they lean economically left and socially liberal or economically right and socially conservative. Revealed ideal points lie on a line, which can be interpreted as

²The paper allows for any log-concave spherical distribution. Section 3 discusses heterogeneous priors.

an endogenous left-right axis. This left-right axis reflects the direction of party disagreement and passes through voters' prior expectation of their ideal points. This result holds for any number of policy issues, assuming a reflection-invariant information cost and a spherical prior.

The mechanism by which cost-effective learning generates ideological polarization is as follows. Voters only need to determine which party they are closer to, not by how much. Thus, they optimally acquire a *binary* signal about their ideal points. This signal sorts voters into two groups: those who believe they are closer to one party and those who believe they are closer to the other. Even centrist voters categorize themselves based on this binary information, causing them to perceive their preferences as more extreme than they truly are.

The second part of this paper examines whether elections produce policies that serve voters' interests when voter positions result from rational learning. To address this, we endogenize party platforms and explore their interaction with voter learning. Parties strategically position themselves in the policy space, balancing electoral prospects against their own policy preferences.

Parties move first and voter learning responds to party positions. This gives parties an agenda-setting role: through their policies, parties influence which issues voters pay attention to. If a party polarizes on a policy issue, voters will pay more attention to this issue as it becomes more relevant to the electoral choice. This timing introduces two novel strategic forces: a *moderation force* and a *polarization force*. First, parties may moderate their policy platforms to bias voter learning in their favor. Second, the more extreme party may find it optimal to polarize, or differentiate from the moderate party to trigger more voter learning, thereby reducing the bias toward their opponent. As information costs decline, the moderation force weakens, leading to increased platform polarization. The polarization force implies that parties may adopt positions more extreme than their own ideal policies to strategically affect voter learning. In contrast, in models with exogenous voter positions (e.g., Roemer 1997), parties never adopt positions more extreme than their own ideal policies.

Our results call for caution when interpreting empirical findings that voter ideology is influenced by party elites as a sign of voter irrationality. Political scientists have long argued that political elites have a large influence on the ideology of voters (Campbell, Converse, Miller, and Stokes 1960; Zaller 1992; Lenz 2012). Such findings have traditionally been interpreted as evidence of voter irrationality. This interpretation is shared by Achen and Bartels (2017), who argue such voter behavior presents a serious threat to democracy. If parties can shape the ideology of voters instead of merely responding to it, it is unclear whether elections produce governments responsive to the preferences of voters. We hope to contribute to this debate by showing that some forms of party influence on voter ideology are consistent with voter rationality and do not preclude that policy is responsive to voters' true preferences. In our model, both issue alignment and polarization of voter ideal points depend on party platforms. First, the alignment of voter ideal points across policy issues is determined by the relative positions of parties. As illustrated in Figure 1, because party *a* is more left and liberal than party *b*, in the resulting voter ideology a left economic position aligns with a liberal social position. Second, more polarized party platforms can result in more polarized voters (see (6)). Although in both cases voters seem to simply follow party positions, our

model shows such effects result from rational voter learning. At the same time, in our equilibrium, party platforms do respond to voters’ true ideal points, namely to the center of their distribution (Theorems 2 and 3). On a higher level, our model illustrates how revealed preferences may differ systematically from true preferences and how they may do so in a context-dependent way.

Our model has policy implications for addressing issue alignment and polarization. Traditional approaches—such as improving political knowledge (Carpini and Keeter 1996) or breaking up echo chambers (Sunstein 2018)—may be ineffective if issue alignment and polarization stem from rational voter learning. Instead, more fundamental changes to voters’ choice sets would be necessary to incentivize multidimensional and non-binary learning. Multidimensional learning could be incentivized through electoral reforms that increase the number of parties, such as transitioning from plurality elections to proportional representation (as suggested by Corollary 1). Non-binary learning could be incentivized through voting or participation mechanisms that elicit the intensity of voter preferences, such as vote trading (Casella 2005) or quadratic voting (Lalley and Weyl 2018).

The paper is organized as follows. The rest of this section discusses related literature and evidence on voter ideology. Section 2 introduces the model. Section 3 analyzes voter learning. Section 4 studies electoral competition when voter ideology results from optimal learning. Section 5 concludes.

Related Literature This paper contributes to the growing literature on rationally inattentive voters. We show rational inattention explains properties of voter ideology by studying *flexible* learning about *ideal points*. Matějka and Tabellini (2021) study electoral competition where voters are inattentive to *party platforms*. They show more attentive voters influence platforms more strongly, as they respond more to them. By contrast, we show attention to *ideal points* reduces responsiveness to platforms, which increases polarization when information becomes cheaper. Yuksel (2022) analyzes voter learning under partitional signals and finds that more specialized learning increases polarization. We allow for flexible learning and show that cheaper information increases polarization through a different mechanism. Li and Hu (2023) study attention to implemented policies in an electoral-accountability setting. They show that the welfare effects of increased attention and mass polarization are ambiguous. Hu, Li, and Segal (2023) analyze learning about valence through an attention-maximizing intermediary. They show such learning generates policy divergence even with office-motivated candidates. By contrast, we study divergence as arising from ideologically motivated candidates and study how it interacts with voter learning.

The literature has proposed several explanations for issue alignment and polarized ideology. Converse (1964) introduced the concept of issue alignment, which he termed ideological constraint, attributing it to logical, psychological, and social sources. Enke, Rodríguez-Padilla, and Zimmermann (2023) propose that moral universalism, the degree of altruism toward strangers versus in-group members, explains correlations in policy views across domains. Spector (2000) derives one-dimensional ideology from cheap talk between two groups with differing priors, while DeMarzo, Vayanos, and Zwiebel (2003) attribute it to networks and a persuasion bias. Bayesian persuasion

can also generate low-dimensional types (Rayo and Segal 2010; Malamud and Schrimpf 2022).³ Ideological polarization has been linked to cognitive limitations, such as correlation neglect (Levy and Razin 2015; Ortoleva and Snowberg 2015) and bounded rationality (Eguia and Hu 2022). Other papers have also proposed rational inattention as a mechanism for belief polarization, albeit through ex-ante heterogeneity in preferences (Novák, Matveenko, and Ravaioli 2024) or path-dependent sequential information acquisition (Nimark and Sundaresan 2019). Callander and Carbajal (2022) explain dynamic polarization through voters adjusting their ideal points toward the party they voted for to rationalize their choice. Our model complements these works by providing a unified explanation for both issue alignment and polarization.

The literature has provided many mechanisms for platform divergence, breaking the median voter result by Downs (1957). We do not propose a new mechanism but show how the new ingredient of our model—endogenous ideology formation through voter learning—interacts with perhaps the most prominent mechanism for platform divergence: ideologically motivated parties and probabilistic voting. Ideological parties still converge to the median voter unless the electoral outcome is uncertain (Wittman 1983; Hansson and Stuart 1984; Calvert 1985). The two common ways to introduce electoral uncertainty are through uncertainty about the ideological position of the median voter (Roemer 1994) and through valence shocks (Hinich 1977; Lindbeck and Weibull 1987). Our model falls into the latter category, which, according to Duggan (2017), has seen little, if any, formal analysis under ideologically motivated parties. We introduce a version of this model that is tractable even in a multidimensional policy space.

1.1 Evidence on Voter Ideology

We discuss the evidence on issue alignment and ideological polarization.

1.1.1 Issue Alignment

Recent evidence shows that the ideology of US voters is approximately one-dimensional (Jessee 2009; Jessee 2012; Tausanovitch and Warshaw 2012; Shor and Rogowski 2018; Fowler, Hill, Lewis, Tausanovitch, Vavreck, and Warshaw 2022; Hare, Highton, and Jones 2023). These studies use voter surveys, such as the American National Election Studies, to estimate voter ideal points, similar to the ideal point estimation of legislators from roll-call data (Poole and Rosenthal 1985). Each study fits a one-dimensional spatial model with quadratic utility to the binary responses that individuals give to policy questions, such as whether the minimum wage should be raised. These studies find that a one-dimensional model already explains voter responses well, accounting for about 80% of binary responses, and that adding dimensions raises explanatory power only marginally. They conclude that voter ideology is well described by a one-dimensional ideological spectrum.

³More broadly related is McMurray (2023), who shows that pivotality considerations in multidimensional common-value elections drive party platforms to bundle logically related issues, reducing platforms to a persistent one-dimensional axis.

In contrast, an older literature starting with Converse (1964) studies the correlation between voter responses to different policy questions rather than estimating ideal points. These papers typically find low correlation and conclude that ideology is not well represented by a one-dimensional spectrum, or that there is little *constraint* on voter ideology in the terminology of Converse. Later research has shown that the low correlation is partly driven by response mistakes, such as those arising from inattentive respondents (Ansolabehere, Rodden and Snyder 2008). The literature seems to have overlooked a further reason why the correlation between responses is a poor measure of how one-dimensional voter ideology is. Even if voters answer according to a one-dimensional spatial model, they do not give left or right answers consistently but pick whichever of a question’s two options (e.g., raising minimum wage or not, expanding military presence abroad or not) lies closer to their ideal point. Each question therefore splits the electorate at a cutoff on the left-right axis, with voters to the right of the cutoff giving the right response and voters to the left giving the left response. Only voters near the extremes fall on the same side of every cutoff and answer consistently, so moderates give a mix of left and right responses even under a fully one-dimensional model.

1.1.2 Polarized Ideology

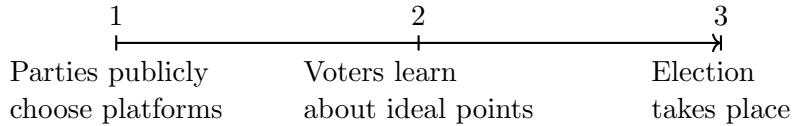
Whether voters are ideologically polarized, that is, have a bimodal distribution, led to an academic debate between Abramowitz and Saunders (e.g., Abramowitz and Saunders 2008) on the affirmative side, and Fiorina, Abrams, Pope, and Levendusky (e.g., Fiorina and Abrams 2008) on the other (see Lelkes (2016) for a critical overview of the debate). Unfortunately, neither side of the debate estimates voter ideal points. Both rely on raw survey evidence, so it is not clear how to interpret their findings. For example, much of the evidence against ideological polarization stems from ideological self-placements on 7-point scales (e.g., Fiorina and Abrams 2008). The validity of ideological self-placements has been criticized for several reasons but the literature has, to our knowledge, overlooked a more fundamental problem. A 7-point scale is a categorical, *ordinal* scale. To assess the bimodality of the distribution of voter ideology, a *cardinal* scale is needed. The reason is that one can always monotonically transform the scale to make a distribution bimodal or unimodal. An ideological 7-point scale is only meaningful if one assumes the 7 categories correspond to intervals of the same size on the appropriate cardinal scale of ideology, for which the authors provide no evidence. This underlines the importance of estimating ideal points from survey responses to multiple questions, as this yields ideal points on a meaningful cardinal scale. On the other hand, the evidence *for* polarization relies on measures of individual-level correlation between left and right responses to different policy questions (Abramowitz and Saunders 2008). However, without estimation of ideal points it is not clear whether their findings relate to issue alignment or polarization of ideology.

Several newer papers do, however, estimate voter ideal points from survey responses. Ba-fumi and Herron (2010) find a bimodal distribution of voter ideal points, while most papers find a unimodal distribution (e.g., Jessee 2012; Hill and Tausanovitch 2015; Dun and Jessee 2020).

However, there are reasons to believe that current estimation algorithms underestimate ideological polarization of voters. Survey respondents who do not pay much attention to the survey are artificially placed in the middle of the distribution, because this best explains their random responses (McCarty 2019, p. 204). Indeed, when Fowler, Hill, Lewis, Tausanovitch, Vavreck, and Warshaw (2022) screen for inattentive respondents (and for respondents that are not well-represented by a one-dimensional ideal point), they find a more bimodal distributions of ideal points in most survey years. Moreover, Abramowitz (2010) finds the distribution of actual or engaged voters is more polarized. We conclude that the matter is not settled yet.

2 Model

Game The policy space is $\mathbb{R}^n$ where each dimension corresponds to a policy issue. Voter ideal points as well as party platforms live in this space. There is a continuum of voters $i \in [0, 1]$ and two parties, a and b . The timing is as follows.



- (1) Parties a and b simultaneously and publicly commit to policy platforms $x_a \in \mathbb{R}^n$ and $x_b \in \mathbb{R}^n$, respectively.
- (2) Each voter $i \in [0, 1]$ learns about her ideal point $\theta_i \in \mathbb{R}^n$.
- (3) Voters vote for their preferred party. The party with the majority wins, and each party wins with probability one half if a tie occurs.

Parties choose platforms $x_a, x_b \in \mathbb{R}^n$ before voters learn about their ideal points. Thus, parties act as agenda-setters: through their platforms, parties affect what issues voters pay attention to. For example, if a party polarizes on migration, then voters will pay more attention to this issue because it becomes more relevant to their voting decision. If, alternatively, voters learned before parties moved, the main result on voter learning (Theorem 1) would remain unchanged. The reason is that the qualitative predictions on voter ideology do not depend on whether voters learn given observed or given anticipated platforms.

Voters The voter utility includes policy utility u and information cost c ,

$$U_i = u(x, \theta_i) - \kappa \cdot c. \quad (1)$$

Following the workhorse model of spatial voting, we assume voters' policy utility is quadratic in the difference between their ideal point and the policy. That is,

$$u(x, \theta) = -(x - \theta)^\top A(x - \theta), \quad (2)$$

where $A \in \mathbb{R}^{n \times n}$ is an arbitrary symmetric, positive definite matrix representing voters' priorities. The evidence on voter ideal points assumes quadratic utility with homogeneous priorities A and shows this specification explains survey responses well (see section 1.1).⁴ Quadratic utility allows us to speak about “revealed ideal points” of voters who have a non-degenerate belief π about their true ideal point θ , as the following remark shows.

Remark 1. *A voter with belief π over her ideal point θ votes as if she had the known ideal point $\mathbb{E}_\pi[\theta]$.*

By a bias-variance decomposition of the expected utility from policy x under belief π over θ ,

$$\mathbb{E}_\pi[u(x, \theta)] = u(x, \mathbb{E}_\pi[\theta]) - \mathbb{E}_\pi[(\theta - \mathbb{E}_\pi[\theta])^\top A(\theta - \mathbb{E}_\pi[\theta])]. \quad (3)$$

The latter term does not depend on x , so when a voter compares two platforms (or survey response items), she chooses the one that is closer to her posterior mean according to the distance induced by A . Hence, a voter with belief π acts like a voter with a known ideal point of $\mathbb{E}_\pi[\theta]$. Accordingly, we call the posterior mean of a voter's belief π over θ her *revealed ideal point*. This ideal point is the one that is estimated from survey responses, which is important when we interpret empirical findings about voter ideology. We refer to the distribution $\rho \in \Delta(\mathbb{R}^n)$ over posterior means induced by learning as the *revealed ideology* in the population, to distinguish it from the true distribution over ideal points.

Learning Voter ideal points $\theta_i \in \mathbb{R}^n$ are distributed according to a distribution $\mu \in \Delta(\mathbb{R}^n)$ with a log-concave and spherical density f_μ and mean normalized to 0.⁵ The density f_μ is spherical if it is of the form $f_\mu(\theta) = g(\theta^\top \theta)$ with $g: \mathbb{R}_+ \rightarrow \mathbb{R}_+$. A spherical density is rotationally symmetric and implies that the ideal points are uncorrelated across dimensions, so there is no issue alignment at the outset. Thus, we allow for a non-parametric class of distributions, examples being the normal distribution $\mathcal{N}(0, k I_n)$ or a uniform distribution on a ball.

Voters share a homogeneous prior μ , conforming to the true distribution. Each voter can acquire *any* signal structure (Blackwell experiment) about her ideal point subject to an information cost, as in the rational-inattention literature (Sims 2003; see also the survey Maćkowiak, Matějka, and Wiederholt 2023). The information cost is the expected reduction in the variance of the voter's belief about her ideal. To define the information cost, recall that a signal structure specifies a conditional distribution over signal realizations given any ideal point. Upon a signal realization, the voter forms a posterior via Bayesian updating. Thus, a signal structure induces a distribution over posteriors. Following the posterior approach, we identify signal structures with their induced distributions over posteriors $\tau \in \Delta(\Delta(\mathbb{R}^n))$. The information cost of distribution τ over posteriors π is then

$$c(\tau) = \text{Var}_\mu(\theta) - \mathbb{E}_\tau[\text{Var}_\pi(\theta)],$$

⁴Moreover, Bachmann, Sarasua, and Bernstein (2024) find that out of a range of commonly used algorithms, the one based on quadratic utility performs best at predicting survey responses.

⁵For any topological space X , we denote by $\Delta(X)$ the set of Borel probability measures on X .

where $\text{Var}_\pi(X) := \mathbb{E}_\pi[\|X - \mathbb{E}_\pi[X]\|^2]$ is the multidimensional analogue of variance. The information cost is non-negative and Blackwell monotone, capturing that learning takes time and effort. Like information costs used in recent contributions (e.g., Ravid, Roesler, and Szentes 2022; Thereze 2022), our information cost belongs to the class of posterior-mean-separable information-costs axiomatized in Mensch and Malik (2024). The information cost is particularly tractable in our setting because it equals the variance of the posterior means by the law of total variance,

$$c(\tau) = \text{Var}_\tau(\mathbb{E}_\pi[\theta]).$$

Together with the quadratic policy utility and expressive voting (see below), this information cost allows us to express the voter’s objective as a simple function of the distribution over posterior means $\rho \in \Delta(\mathbb{R}^n)$, induced by τ , as we detail in the Appendix:

$$\begin{aligned} \max_{\rho \in \Delta(\mathbb{R}^n)} \mathbb{E}_{\theta \sim \rho} \left[\left| (x_b - x_a)^\top A \left(\theta - \frac{x_a + x_b}{2} \right) \right| - \kappa \theta^\top \theta \right] \\ \text{s.t. } \rho \preceq_{\text{MPS}} \mu. \end{aligned} \quad (4)$$

Here, we use that a distribution $\rho \in \Delta(\mathbb{R}^n)$ over posterior means is induced by some signal structure if and only if ρ is a mean-preserving contraction of the prior μ , $\rho \preceq_{\text{MPS}} \mu$.⁶ Due to the simple form of this objective, consisting of a piecewise linear and a quadratic function, we can obtain a closed-form solution for the voter’s learning problem in some parameter region. While the qualitative results on voter learning hold for a more general class of information costs, this cost function proves particularly tractable to solve for endogenous party positions.

For voters to acquire costly information despite never being pivotal among the continuum of voters, we assume voters engage in *expressive voting*, as is standard in the literature on rationally inattentive voters (Matějka and Tabellini 2021; Hu, Li, and Segal 2023; Li and Hu 2023). That is, voters genuinely care about voting for the correct candidate given their true preferences, for which they are willing to incur an information cost. The reason may be that voters derive a psychological benefit from doing so or they consider it their civic duty (see also Feddersen and Sandroni 2006).

Parties We assume that parties are policy-motivated (Wittman 1973; Calvert 1985). The utilities U_a and U_b of parties a and b , respectively, are

$$\begin{aligned} U_a(x_a, x_b) &= P_a(x_a, x_b)u(x_a, x_a^*) + (1 - P_a(x_a, x_b))u(x_b, x_a^*) \\ U_b(x_a, x_b) &= P_a(x_a, x_b)u(x_a, x_b^*) + (1 - P_a(x_a, x_b))u(x_b, x_b^*), \end{aligned}$$

where $P_a(x_a, x_b)$ is the probability that party a ’s policy x_a gets implemented. We make the common assumption that this implementation probability is the vote share (Wittman 1983; Wittman 1990; Callander and Carbajal 2022; Yuksel 2022). A continuous mapping from the vote share to the implementation probability can result from additional noise voters or random turnout of partisans

⁶A distribution $\rho \in \Delta(\mathbb{R}^n)$ is a mean-preserving contraction of another distribution $\mu \in \Delta(\mathbb{R}^n)$ if there exist $\mathbb{R}^n$ -valued random variables $X \sim \rho, Y \sim \mu$ with $\mathbb{E}[Y|X] = X$.

(e.g., Feddersen and Pesendorfer 1996), or from non-majoritarian institutions in parliament.⁷

Equilibrium We focus on subgame perfect pure-strategy equilibria, in which both parties receive positive vote shares. In such an equilibrium, voters vote for their preferred party at the election stage, voters learn optimally given observed party platforms, and parties choose their platforms optimally anticipating voter behavior. Equilibria where one party obtains all the votes are not robust to perturbations such as introducing an arbitrarily small office benefit.

2.1 Discussion of Assumptions

Learning about Ideal Points Voter learning about ideal points can be interpreted as (i) introspecting on how to *value* the consequences of policies, (ii) learning about the private *consequences* of policies (recall ideal points are private), or (iii) a combination of both. As an example, consider a voter’s position on income taxation. To determine her optimal tax policy, the voter may want to introspect on her values for equity versus efficiency, as well as learn about what tax bracket she is in and what other economic consequences the policy has. We remain agnostic as to which interpretation should be adopted.

We assume voters can acquire costly information about their ideal points, but they observe party platforms for free. This approach allows us to make clear what mechanisms result from endogenous voter learning about ideal points, as opposed to learning about platforms or valence (for the latter, see Matějka and Tabellini 2021; Hu, Li, and Segal 2023). However, as we briefly illustrate in Appendix C.1, our results on voter ideology also hold when voters do not know platforms but observe a common signal about platforms, based on which they choose how to learn about ideal points. What is more importantly ruled out by our assumption is that voters learn *jointly* about platforms and ideal points, which is an interesting avenue for future research.

Flexible Information Acquisition The rational-inattention approach preserves tractability while allowing complete flexibility in what kind of information voters can acquire. The flexibility assumption ensures the optimal signal structure is determined endogenously and not through exogenous restrictions. In particular, we are not imposing that signals about different policy issues need to be independent. That is, voters can, for example, acquire a signal that informs them about whether they are left or right, when aggregating their positions on multiple policy issues. We show such signals are, in fact, optimal.

The substantive meaning of this assumption depends on which of the above-mentioned interpretations of voter learning we adopt. When we interpret voter learning as learning about values, one can think of the voter imagining two policies that differ on multiple issues and introspecting on their relative desirability, similar to drift-diffusion models, widely used in psychology and neuroscience. By not imposing any restrictions on information, our approach stays true to the original motivation of rational inattention as modeling the brain as an efficient information processor subject to only information capacity constraints (Sims 2003). On the other hand, when we interpret voter learning

⁷As Yuksel (2022) notes, if half the population consists of noise voters and their vote share for party a is uniformly distributed, then the implementation probability is exactly the vote share from non-noise voters.

as learning about private consequences of policies, another way for such learning to be aggregated across dimensions is through information intermediaries, as in Hu, Li, and Segal (2023). Voters may learn about private policy consequences from sufficiently personalized media outlets, such as news feeds or newspapers catering to specific demographics. Such media outlets may aggregate information about different policy issues to a one-dimensional signal, as other models of media assume (Duggan and Martinelli 2011; Yuksel 2022; Perego and Yuksel 2022).

3 Voter Learning

This section characterizes optimal voter learning given party platforms x_a and x_b , assuming $x_a \neq x_b$ (otherwise, voters learn nothing). The next section endogenizes the platforms. This section can be read independently of the political-economy application: the results apply analogously to an industrial-organization setting with horizontally differentiated goods in a product-attribute space.

The following lemma shows by standard arguments that a solution to the voter learning problem exists and is unique.

Lemma 1. *The voter learning problem (4) has a unique solution.*

The main result of this section describes the revealed ideal points (posterior means) that are induced by the optimal voter learning strategy.

Theorem 1. *The solution to the voter learning problem (4) is supported on at most two revealed ideal points on the line through the prior mean with direction $A(x_b - x_a)$.*

Figure 2 visualizes the result. Even though the true distribution of ideal points is multidimensional and unimodal, the revealed ideology in the population is one-dimensional and binary. By implication, the revealed ideal points are perfectly aligned across policy issues, even if the true ideal points are independent across dimensions. Because the line does not depend on κ , revealed ideal points lie on it even when voters are heterogeneous in their information cost. We interpret the line as an *endogenous left-right axis* of voter ideology. In the Appendix, we show that revealed ideal points lying on a line is exactly the property that makes predicted survey behavior indistinguishable from that under a one-dimensional political space (Proposition 5), as used in much of formal political economy. Heterogeneity in κ smooths out revealed voter ideology, but if the distribution of κ is sufficiently concentrated, voter ideology will be bimodal, that is, polarized.

On a high level, the intuition for the result is that voters learn only about decision-relevant information, which leads them to ignore all but one dimension of the policy space and to learn in a binary fashion. This is familiar from the rational-inattention literature, where agents ignore payoff-irrelevant dimensions and acquire no more signals than there are actions. Surprisingly, the connection to issue alignment and polarization has gone unnoticed, and the literature has pursued more complicated explanations instead. Having said that, the alignment result does not follow mechanically from a standard rational-inattention argument. Instead, the proof rests on a novel reflection argument that exploits symmetries of the decision problem. Importantly, the argument

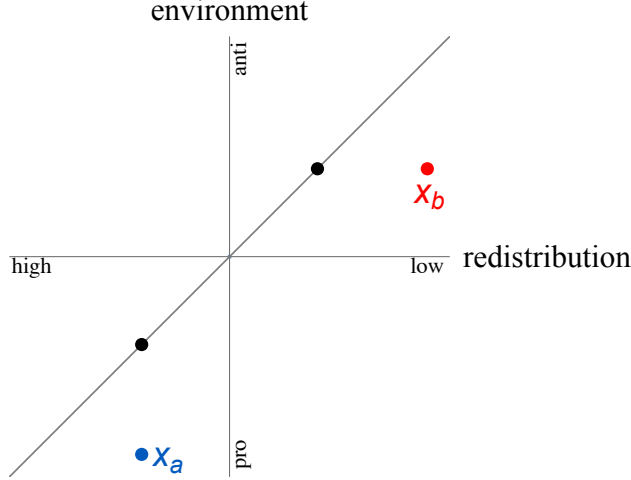


Figure 2: Revealed ideal points (•) under party platforms x_a, x_b and priorities $A = I_n$.

does not rely on the controversial invariance property of mutual information (Dean and Neligh 2023).

The proof uses that in the posterior-mean reformulation of the voter’s objective, (4), the instrumental value depends on a posterior mean θ only through its projection onto the line L through the prior mean with direction $A(x_b - x_a)$, whereas the cost depends only on the squared distance $\theta^\top \theta$ to the prior mean. Both are therefore invariant under reflection across L , which preserves both the projection onto L and the distance to the prior mean.

Suppose by way of contradiction that voters acquired information such that the induced distribution ρ over posterior means was not supported on at most two points on the line L (Figure 3(a)). We construct in four steps a distribution with the same instrumental value but a lower cost. First, the voter is indifferent between ρ and the distribution obtained by reflecting ρ across L (Figure 3(b)): this reflection is Bayes-consistent because the prior is spherical, and by the above it leaves both value and cost unchanged. Second, because the information cost is posterior separable, the voter is indifferent to mixing these two equivalent distributions, and hence to acquiring the mixture $\frac{1}{2}\rho + \frac{1}{2}\text{ref}(\rho)$ (Figure 3(c)) instead. This mixture is symmetric about L . Third, this symmetry implies that projecting the mixture orthogonally onto L is a mean-preserving contraction (Figure 3(d)). This mean-preserving contraction corresponds to a garbling that reduces the information cost while preserving the instrumental value, since it maintains the projection on L . Finally, garbling once more according to the action recommendation yields a binary distribution with the same instrumental value and a further reduced information cost (Figure 3(e)). The argument uses only reflection invariance, posterior separability, and Blackwell monotonicity of the information cost.

Alignment survives when voters also want to learn how strongly they prefer one party over the other. Suppose voters care about the intensity of their preference, for instance because they vote only when the expected utility difference between the parties is large enough and abstain otherwise. The first three steps of the proof go through unchanged, so revealed ideal points still lie on the line with direction $A(x_b - x_a)$. The binarization step no longer applies. Alongside the two groups, a

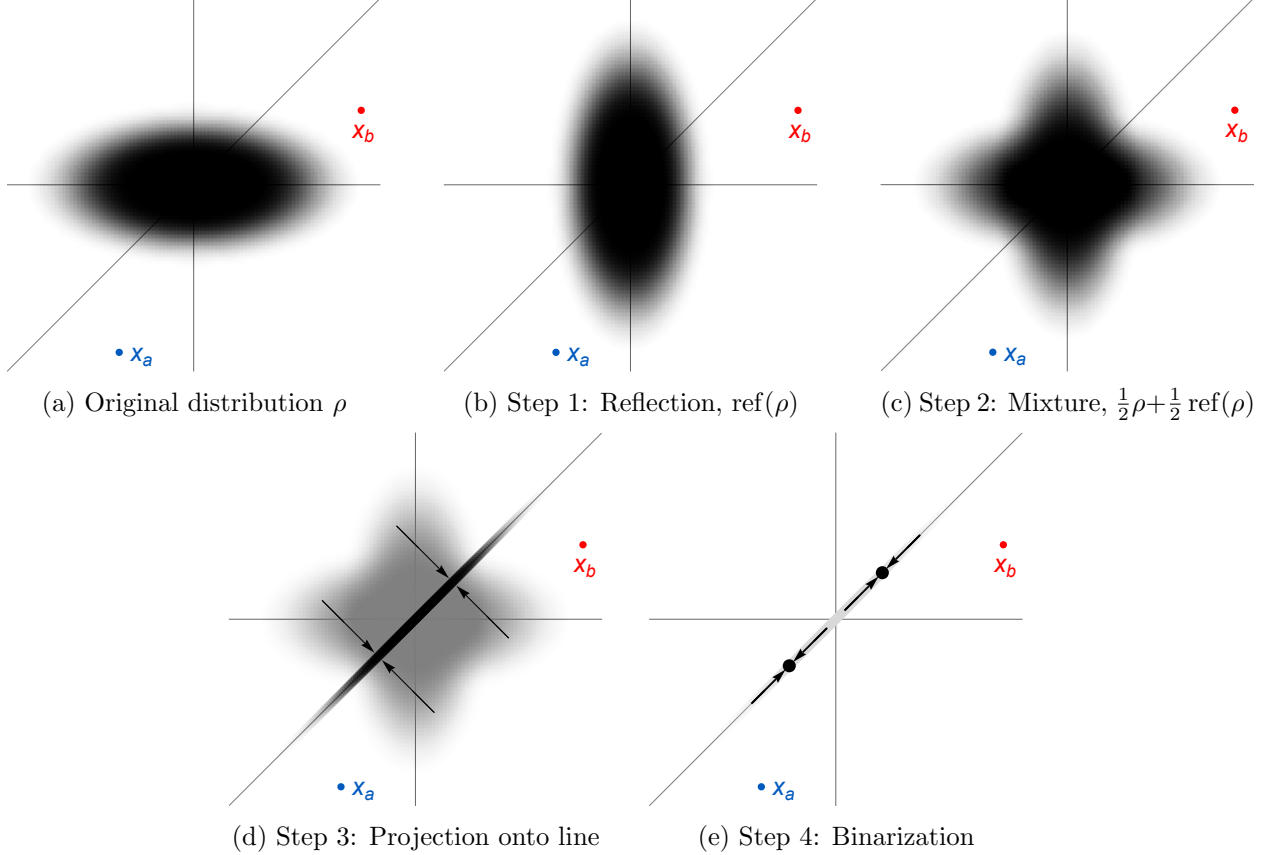


Figure 3: Reflection argument underlying the proof of Theorem 1 for $A = I_n$.

third emerges, which we can interpret as disengaged voters who do not feel strongly enough about either party to turn out.

The line with direction $A(x_b - x_a)$ also organizes voter ideology when prior means are heterogeneous. Suppose voters share a common prior $\theta_i \sim \mathcal{N}(0, I_n)$ and each observes an exogenous signal $S_i = \theta_i + \varepsilon_i$ with $\varepsilon_i \sim \mathcal{N}(0, \sigma^2 I_n)$ independent of θ_i before learning. Interim beliefs are then spherical with heterogeneous means. Each voter learns along the line through her own prior mean with direction $A(x_b - x_a)$, and the binary support induced by learning is spread along that direction. The marginal distribution of revealed ideal points orthogonal to $A(x_b - x_a)$ is unaffected by learning, while the marginal along $A(x_b - x_a)$ undergoes a mean-preserving spread through learning. When the spread from learning is large relative to the dispersion of prior means, the distribution of revealed ideal points is concentrated near the direction $A(x_b - x_a)$, so voter ideal points are approximately aligned.

Theorem 1 relies on there being only two parties, but an appropriate extension survives with more parties. Under plurality rule, in a Duvergerian equilibrium where voters decide between the two front-runners, the mechanism still applies.⁸ Under proportional representation, expressive voting may remain a reasonable assumption. With $k > 2$ parties an analogous reflection argument

⁸Although we assume expressive voting, strategic voting does not undermine the result under private values as long as voters perceive a positive probability of being pivotal.

shows that revealed ideal points lie on a hyperplane of dimension at most $k - 1$. If the k platforms happen to lie on a line, voter ideology is still one-dimensional. If party platforms lie in general position, the dimensionality of voter ideology rises with the number of relevant parties.

Corollary 1. *Suppose voters face k party platforms $x_1, \dots, x_k$ and choose a Bayes-consistent $\tau \in \Delta(\Delta(\mathbb{R}^n))$ to maximize $\mathbb{E}_\tau[\max_{j=1, \dots, k} \mathbb{E}_\pi[u(x_j, \theta)]] - \kappa c(\tau)$. Then any solution is supported on at most k revealed ideal points, contained in the subspace through the prior mean spanned by $\{A(x_j - x_1)\}_{j=2, \dots, k}$ with dimension at most $k - 1$.*

This result aligns with evidence from Western European countries with more than two relevant parties. For these countries a two-dimensional space is needed to describe party and voter positions (Kriesi, Grande, Lachat, Dolezal, Bornschier, and Frey 2006; Kriesi, Grande, Lachat, Dolezal, Bornschier, and Frey 2008; Bornschier 2010a; Bornschier 2010b; however, see Van Der Brug and Van Spanje 2009).

3.1 Vote Shares

The vote shares induced by voter learning, and the bias established below, are the central inputs to the analysis of electoral competition in the next section. Each vote share is the weight that the optimal revealed ideology places on one of the two revealed ideal points. Under party positions (x_a, x_b) , let $\rho(x_a, x_b)$ be the unique optimal revealed voter ideology. The vote share of party a is

$$P_a(x_a, x_b) = \mathbb{E}_{\rho(x_a, x_b)}[\mathbb{1}((\theta - x_a)^\top A(\theta - x_a) \leq (\theta - x_b)^\top A(\theta - x_b))],$$

the mass of voters whose revealed ideal point is A -closer to x_a than to x_b , and the vote share of party b is $1 - P_a$. By Theorem 1 the optimal ρ is at most binary, so $P_a(x_a, x_b)$ is the weight ρ places on the revealed ideal point that is A -closer to x_a . We assume voters break indifference in favor of party a but this assumption is immaterial because the optimal ρ places no mass on the indifference set if $x_a \neq x_b$.

The vote shares are in general different from the share of voters whose true ideal point is A -closer to x_a than to x_b , given by the same indicator under the prior μ ,

$$\mathbb{E}_\mu[\mathbb{1}((\theta - x_a)^\top A(\theta - x_a) \leq (\theta - x_b)^\top A(\theta - x_b))].$$

The next result shows that costly learning biases the vote share toward the moderate party, the party whose position is A -closer to the prior mean 0. The moderate party receives a larger vote share than the share of voters whose true ideal point is closer to its position, even though voters update beliefs in an unbiased, Bayesian manner.

Proposition 1. *Suppose $x_a^\top A x_a < x_b^\top A x_b$. Then party a 's vote share $P_a(x_a, x_b)$ weakly exceeds the share of voters whose true ideal point is A -closer to x_a than to x_b . The vote share $P_a(x_a, x_b)$ weakly increases if we raise the information cost parameter κ , and weakly decreases if we scale up $x_b - x_a$ while holding the midpoint $\frac{x_a + x_b}{2}$ fixed.*

The high-level reason for the bias is that asymmetric signals are less informative and hence cheaper. Voters acquire signals biased toward the moderate party to economize on their information cost. The extreme case is immediate: when information is prohibitively costly, voters learn nothing and all vote for the moderate party. The proposition shows that the bias persists at every finite cost. The bias is smaller when information is cheaper or when parties are more polarized, because in those cases the motive to economize on information is weaker.

Under rational inattention, behavior can be biased toward ex-ante more attractive options (Matějka and McKay 2015). Precise statements outside the two-state, two-action case are not available, to our knowledge, because the unconditional choice probabilities are hard to characterize under a mutual-information cost. Our setting, with quadratic utility and a posterior-variance cost, is more tractable. The key is that the voter learning problem reduces to a one-dimensional mean-measurable information design problem, which is well understood.

To unpack the intuition in the present setup, consider the signal structures that implement the distribution of revealed ideal points as κ rises. By Theorem 1 this distribution is one-dimensional and it can be implemented by a signal that informs voters only about the scalar projection of their ideal point onto the axis $A(x_b - x_a)$. The discussion below therefore concerns signal structures that are functions of this scalar projection. Voters are indifferent between the two parties at the scalar projection of $\frac{x_a + x_b}{2}$, which we call the party midpoint. When κ is small, voters acquire a threshold signal, with the threshold near the party midpoint. The signal almost perfectly separates voters according to which party their true ideal point is closer to, so vote shares approximate the true shares. As κ increases, the optimal threshold becomes more extreme than the party midpoint and moves toward the extreme party. This happens because, under the log-concave prior, a more extreme threshold reduces the posterior-variance information cost. A threshold past the party midpoint assigns more voters to the moderate party, and this is the source of the bias. As κ rises further, voters shift from adjusting the threshold to garbling the binary signal. For these values of κ , the two revealed ideal points are symmetric about the party midpoint, which equalizes the cost of a mistaken vote for each party from garbling. Higher κ contracts these two points toward $\frac{x_a + x_b}{2}$ while preserving this symmetry. The vote shares are exactly the weights that make the two revealed points average to the prior mean. Since the prior mean lies on the moderate party's side of the party midpoint, contracting the points toward the party midpoint raises the vote share for the moderate party. When the cost is prohibitive, voters acquire no information and all vote for the moderate party.

Our notion of bias toward the moderate party matches the definition of *media bias* in Gentzkow, Shapiro, and Stone (2015), so the model illustrates a rational-inattention-based rationale for demand-driven media bias. Recall that one interpretation of voter learning is that information is supplied by an intermediary such as the news media. Because the moderate party is defined as the one closer to the voter's prior, the bias favors prior-aligned news. Most demand-driven explanations of media bias rest on a psychological utility from news. We instead show that media bias can be a

rational way for voters to economize on their information cost.⁹ The two channels behind this bias map onto the two forms of media bias in Gentzkow, Shapiro, and Stone (2015): a more extreme threshold is a filtering bias and asymmetric garbling adds a distortion bias.

Proposition 1 is central for electoral competition. Endogenous information creates two new forces on party positions, a **moderation force** and a **polarization force**. First, the bias toward the moderate party gives parties an additional motive to moderate, and this motive strengthens as information becomes more costly. Second, the extreme party has an incentive to polarize, since greater polarization reduces the bias toward the moderate party. The same logic gives the moderate party an incentive to move toward the extreme party, since proximity raises the bias in its own favor. The next section shows that the moderation force is the key force at play in symmetric equilibria and the polarization force is central for understanding asymmetric equilibria.

4 Electoral Competition

Solving for the equilibrium party positions is challenging because of the multidimensional policy space and endogenous voter learning. Existence and equilibrium characterizations with policy-motivated parties are confined to one-dimensional policy spaces in which the median voter is pivotal (Duggan 2017). Our model adds endogenous voter learning, which interacts with a further difficulty of electoral-competition models: party objectives need not be quasi-concave, so pure-strategy equilibria may not exist.

For this section, we set uniform priorities, $A = I_n$, for tractability. Then particularly simple closed-form solutions for the revealed voter ideal points and the vote shares hold. To define them, let $\text{Proj}_{y,z}$ denote the orthogonal projection on the line through $y, z \in \mathbb{R}^n$,

$$\begin{aligned} \text{Proj}_{y,z} : \mathbb{R}^n &\rightarrow \mathbb{R}^n \\ x &\mapsto y + \frac{(z - y)^\top (x - y)}{\|z - y\|^2} (z - y). \end{aligned} \tag{5}$$

When the prior is dispersed enough, the two revealed ideal points $\theta_a, \theta_b \in \mathbb{R}^n$ given party positions $x_a, x_b \in \mathbb{R}^n$ are

$$\begin{aligned} \theta_a(x_a, x_b) &:= \text{Proj}_{0, x_b - x_a} \left(\frac{x_a + x_b}{2} \right) - \frac{x_b - x_a}{2\kappa} \\ \theta_b(x_a, x_b) &:= \text{Proj}_{0, x_b - x_a} \left(\frac{x_a + x_b}{2} \right) + \frac{x_b - x_a}{2\kappa} \end{aligned} \tag{6}$$

Both revealed ideal points lie on the line through the origin with direction $x_b - x_a$. The first term of (6) is the orthogonal projection of the party midpoint $\frac{x_a + x_b}{2}$ on that line, and θ_a and θ_b are

⁹Our result complements Che and Mierendorff (2019). In their dynamic model voters choose between left- and right-biased news under a time cost, and voters with extreme beliefs acquire news biased toward their current belief. Our model allows flexible information acquisition, so voters may acquire unbiased news, yet the information they choose is always biased toward the party closer to their prior belief.

equally far from it. These solutions imply a simple expression for the vote share,

$$P_a(x_a, x_b) = \frac{1}{2} + \frac{\kappa}{2} \frac{\|x_b\|^2 - \|x_a\|^2}{\|x_b - x_a\|^2}.$$

However, these closed-form solutions hold only when the revealed ideal points θ_a and θ_b are feasible, that is, when some distribution on $\{\theta_a, \theta_b\}$ is a mean-preserving contraction of the voter prior μ . The closed-form solutions therefore do not by themselves deliver equilibrium. Even if feasibility does not bind at given party positions, establishing a global best response requires ruling out deviations into the region where the feasibility constraint binds and the closed forms fail.

To characterize equilibrium despite this challenge, our multidimensional best response lemma reduces the electoral competition game to one dimension. It is useful to split a party's best response into two choices: the direction in which it differentiates from its opponent, and, given that direction, the extent of differentiation. Fixing the direction restricts the two positions to a line. Along that line we show that the multidimensional party game is equivalent to a one-dimensional party game in which the voter prior and the party ideal points are projected onto it. This one-dimensional game is tractable because the voter learning problem reduces to a one-dimensional mean-measurable persuasion problem, which is well understood. The resulting solution to the voter problem makes a party's best response affine in its opponent's position wherever feasibility does not bind. We use this to solve for the optimal direction: party j 's best response direction passes through $(1 - \kappa)x_j^*$. It follows that in equilibrium both parties lie on the line through $(1 - \kappa)x_a^*$ and $(1 - \kappa)x_b^*$, and that their positions along it coincide with those of the corresponding one-dimensional model.

The remainder of this section characterizes equilibrium, first under a symmetric setup in which parties are equally ideologically extreme, then under an asymmetric setup in which one party is more ideologically extreme. Two forces shape the equilibrium. A moderation force pulls parties toward the center of the voter distribution, and a polarization force pushes them apart. The symmetric setup isolates the first. The asymmetric setup brings in the second.

4.1 Symmetric Setup

We first characterize the equilibrium under a symmetric setup, that is, when parties are equally ideologically extreme from the viewpoint of voters, $\|x_a^*\| = \|x_b^*\|$. For the statement of the theorem, define $\underline{\kappa}$ via

$$\frac{1 - \underline{\kappa}}{\underline{\kappa}} \frac{\|x_b^* - x_a^*\|}{2} = \mathbb{E}[\|\theta_1\|].$$

where θ_1 denotes the first component of θ . The parameter $\underline{\kappa}$ is the smallest information cost parameter for which the closed-form solution (6) for the voter learning problem still holds. Throughout, δ_x denotes the Dirac measure placing unit mass at x .

Theorem 2. *Let $\|x_a^*\| = \|x_b^*\|$ and $x_a^* \neq x_b^*$. Define $\theta_j(\kappa) := \text{Proj}_{0, x_b^* - x_a^*}((1 - \kappa)x_j^*/\kappa)$ for $j \in \{a, b\}$.*

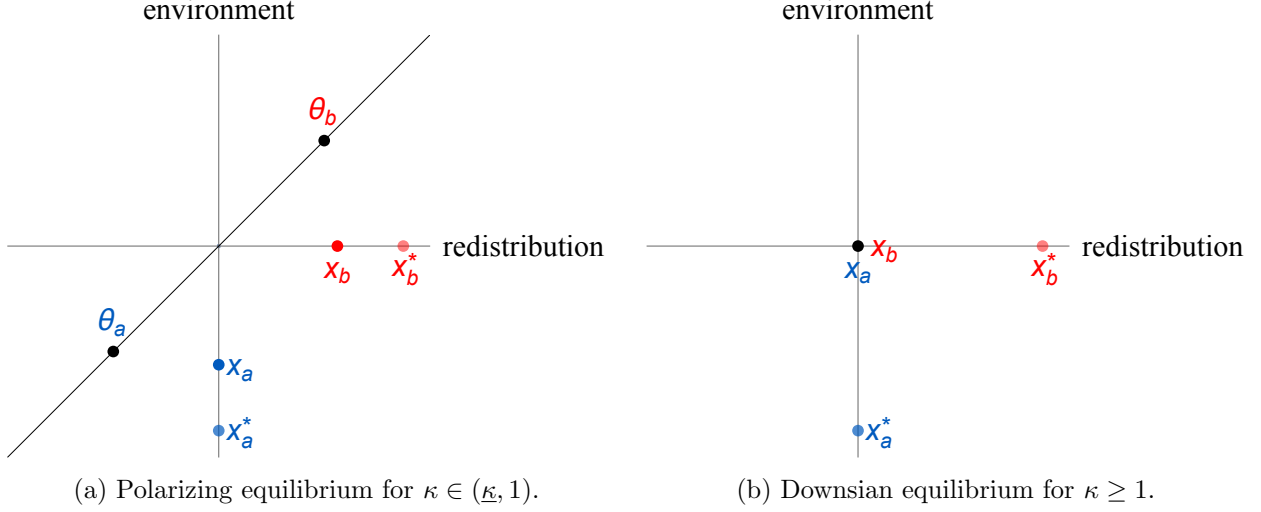


Figure 4: Equilibria under symmetric setup.

The following describes an equilibrium in two parameter regimes.

	Party positions	Revealed voter ideology
$\kappa \in (\underline{\kappa}, 1)$: Polarizing equilibrium	$(x_a, x_b) = (1 - \kappa)(x_a^*, x_b^*)$	$\rho = \frac{1}{2}\delta_{\theta_a(\kappa)} + \frac{1}{2}\delta_{\theta_b(\kappa)}$
$\kappa \geq 1$: Downsian equilibrium	$(x_a, x_b) = (0, 0)$	$\rho = \delta_0$

The equilibrium party positions are unique for $\kappa \geq 1$. For $\kappa \in (\underline{\kappa}, 1)$ with $\kappa \neq 1/2$, the equilibrium party positions are unique if $\|x_b^* - x_a^*\| \leq \kappa \mathbb{E}[|\theta_1|]$.

The theorem describes two types of equilibria, depending on the information cost parameter, displayed in Figure 4.

For large information costs, there is a *Downsian equilibrium*, reminiscent of the median-voter theorem (Downs 1957). In this equilibrium, parties fully converge at the voter's prior expectation of the optimal policy. Voters acquire no information and split their votes arbitrarily between the two parties.

For smaller information costs, there is what we call a *polarizing equilibrium*. In this equilibrium, parties diverge and voters acquire information, which polarizes the population into two groups. Surprisingly, the model allows for a simple closed-form solution of the party and voter positions. The smaller the information cost, the more parties and voters polarize.

The comparative statics are driven by the novel **moderation motive** for parties. For a high-level intuition, recall Corollary 4: to economize on the cost of information, voter learning is biased toward the moderate party. This bias encourages parties to moderate. As information becomes cheaper, the motive to reduce the information cost weakens, decreasing the bias and leading to more party polarization.

To explain the intuition in more detail, let us first recall parties' trade-off when choosing positions. Policy-motivated parties trade off votes (winning often) and ideology (implementing a favorable policy when winning). Parties gain votes by moving closer to voters' prior expectation.

Parties gain ideological utility by moving closer to their own ideal policies, creating polarization. The optimal party positions equate the loss of votes with the gain in ideological utility from polarizing further. As a consequence, the smaller the loss of votes, the larger is the equilibrium level of polarization. Now, the bias toward the moderate party creates an additional loss of votes from polarizing further than the other party, dampening polarization. However, this bias toward the moderate party recedes as information becomes cheaper, creating more party polarization.

A natural conjecture is that cheaper information benefits voters, both directly through a lower cost of learning and indirectly through better-informed voting. The next result shows the opposite in the polarizing regime: cheaper information reduces aggregate voter welfare. We define aggregate voter welfare W under party positions (x_a, x_b) and revealed voter ideology ρ as the total policy utility minus the total information cost:

$$W(x_a, x_b, \rho) = P_a(x_a, x_b)\mathbb{E}[u(x_a, \theta)] + (1 - P_a(x_a, x_b))\mathbb{E}[u(x_b, \theta)] - \kappa c(\rho).$$

We later comment on how the welfare comparative static would change if we included expressive utility from voting for one's preferred party in voter welfare.

Corollary 2. *Let $\|x_a^*\| = \|x_b^*\|$ and $x_a^* \neq x_b^*$. On $\kappa \in (\underline{\kappa}, 1]$, equilibrium aggregate voter welfare W is strictly increasing in κ .*

The intuition has two parts. First, lower κ increases party polarization, moving implemented policies further from the welfare-maximizing policy at the center of the voter distribution, the origin. Second, one might expect voters to reduce their information cost as the information becomes cheaper. However, cheaper information and greater platform polarization make voters acquire more information, and the increase in information acquisition overturns the direct benefit of a smaller information cost parameter.

Under expressive utility from voting for the preferred party, platform polarization can be beneficial by giving voters a choice to vote for the closer party. The mechanism resembles aggregate uncertainty, where platform polarization makes policy responsive to aggregate shocks and can raise welfare (Bernhardt, Duggan and Squintani 2009). Party polarization, however, does more than move the positions further apart, which carries the responsiveness and expressive benefit. Party polarization also moves the positions toward the midpoint of the parties' ideal points $\frac{x_a^* + x_b^*}{2}$, which biases policy away from the origin and lowers welfare. Furthermore, the two negative effects above, on policy utility and information cost, remain.

4.2 Asymmetric Setup

We now turn to the equilibrium when one party is more ideologically extreme, which without loss of generality we take to be party b , so that $\|x_a^*\| < \|x_b^*\|$.

Even under asymmetry, the equilibrium is in closed form for a large parameter region. Define

the party positions

$$\begin{aligned} x_a(\kappa) &:= \text{Proj}_{(1-\kappa)x_a^*, (1-\kappa)x_b^*} \left(\frac{1-\kappa}{1-2\kappa} ((1-\kappa)x_a^* + \kappa x_b^*) \right) \\ x_b(\kappa) &:= \text{Proj}_{(1-\kappa)x_a^*, (1-\kappa)x_b^*} \left(\frac{1-\kappa}{1-2\kappa} ((1-\kappa)x_b^* + \kappa x_a^*) \right), \end{aligned} \quad (7)$$

the revealed ideal points

$$\begin{aligned} \theta_a(\kappa) &:= \text{Proj}_{0, x_b^* - x_a^*} \left(\frac{1-\kappa}{1-2\kappa} \frac{x_a^* + x_b^*}{2} \right) - \frac{1-\kappa}{2\kappa} (x_b^* - x_a^*) \\ \theta_b(\kappa) &:= \text{Proj}_{0, x_b^* - x_a^*} \left(\frac{1-\kappa}{1-2\kappa} \frac{x_a^* + x_b^*}{2} \right) + \frac{1-\kappa}{2\kappa} (x_b^* - x_a^*), \end{aligned}$$

and party a 's vote share

$$P_a(\kappa) := \frac{1}{2} + \frac{\kappa}{2(1-2\kappa)} \frac{\|x_b^*\|^2 - \|x_a^*\|^2}{\|x_b^* - x_a^*\|^2}.$$

The characterization covers the case in which the two parties are ideologically on opposite sides of the center of the voter distribution. Let $\text{proj}_v(x) := \frac{v^\top x}{\|v\|}$ denote the scalar projection onto $v \in \mathbb{R}^n$, and set $\check{x}_a^* := \text{proj}_{x_b^* - x_a^*}(x_a^*)$ and $\check{x}_b^* := \text{proj}_{x_b^* - x_a^*}(x_b^*)$. The relevant case is $\check{x}_a^* < 0 < \check{x}_b^*$, in which the ideal policies project onto opposite sides of the origin along the ideological difference $x_b^* - x_a^*$.

Define the set of information cost parameters

$$\mathcal{K} = \left\{ \kappa \in \left(0, \frac{\check{x}_b^* - \check{x}_a^*}{3\check{x}_b^* - \check{x}_a^*} \right) \cup \left(\frac{\check{x}_b^* - \check{x}_a^*}{\check{x}_b^* - 3\check{x}_a^*}, 1 \right) \mid \max\{\|x_a(\kappa)\|, \|x_b(\kappa)\|\} \leq \kappa \mathbb{E}[\|\theta_1\|] \right\}.$$

Because $\check{x}_a^* + \check{x}_b^* > 0$ under $\|x_a^*\| < \|x_b^*\|$, the value $1/2$ falls strictly between the two intervals.

Theorem 3. *Suppose $\|x_a^*\| < \|x_b^*\|$ and $\check{x}_a^* < 0 < \check{x}_b^*$. The following describes an equilibrium in two parameter regimes.*

	Party positions	Revealed voter ideology
$\kappa \in \mathcal{K}$:	Polarizing equilibrium $(x_a, x_b) = (x_a(\kappa), x_b(\kappa))$	$\rho = P_a(\kappa) \delta_{\theta_a(\kappa)} + (1 - P_a(\kappa)) \delta_{\theta_b(\kappa)}$
$\kappa \geq 1$:	Downsian equilibrium $(x_a, x_b) = (0, 0)$	$\rho = \delta_0$

The equilibrium party positions are unique for $\kappa \geq 1$, and for $\kappa \in \mathcal{K}$ they are unique whenever $\|x_b^ - x_a^*\| \leq \kappa \mathbb{E}[\|\theta_1\|]$.*

Theorem 3 states that in equilibrium, party positions x_a and x_b are on the line through $(1-\kappa)x_a^*$ and $(1-\kappa)x_b^*$, for all $\kappa \in \mathcal{K}$. The positions of parties on said line are given by (7). As in the equilibrium under a symmetric setup, party polarization is given by the simple formula

$$x_b(\kappa) - x_a(\kappa) = (1-\kappa)(x_b^* - x_a^*).$$

In particular, party polarization increases as the information cost κ decreases.

Figure 5 plots the equilibrium positions (x_a, x_b) as a function of κ under a one-dimensional

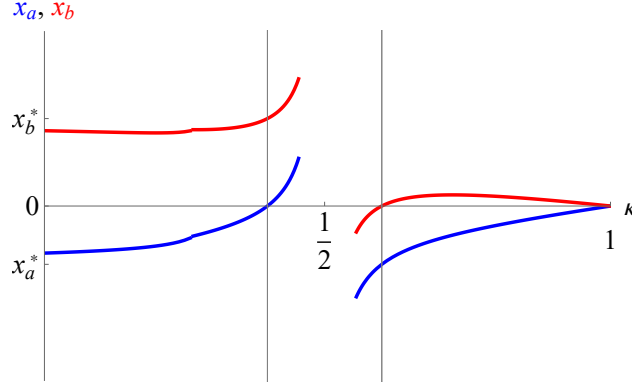


Figure 5: Party positions as a function of the information cost parameter κ

policy space. For large information costs, $\kappa \geq 1$, parties converge fully as in the symmetric setup. For smaller information costs, $\kappa < 1$, parties polarize as κ decreases. For an intermediate range of κ , there is no equilibrium in pure strategies where both parties obtain positive vote shares. We comment on this below. For small κ , Figure 5 plots numerically solved equilibrium positions.

A notable feature of the equilibrium is that for intermediate information costs, both parties are on the same side of zero and one party is more extreme than its ideal policy. This contrasts with the Calvert-Wittman model under exogenous voter positions, where party positions necessarily lie between their ideal policies (Roemer 1997).

This feature of the equilibrium is driven by the **polarization motive**, which creates *chase-and-evade incentives*, similar to the literature on valence advantage (Groseclose 2001; Aragones and Palfrey 2002; Bernhardt, Buisseret, and Hidir 2020). By the logic of the polarization motive, in an asymmetric equilibrium, the extreme party has an incentive to differentiate (“evade”) and the moderate party has an incentive to move closer to (“chase”) the extreme party. When the information cost parameter κ is close to $1/2$, the polarization motive becomes strong enough to push the extreme party beyond its own ideal policy. In other words, endogenous voter learning gives parties an incentive to strategically extremize. The moderate party, on the other hand, moves so close to the extreme party that it is pulled beyond the center of the voter distribution, zero. If κ is very close to $1/2$, these chase-and-evade incentives become so strong that a pure-strategy equilibrium fails to exist, as is common in the literature on valence advantage (Aragones and Palfrey 2002; Hummel 2010).

5 Conclusion

In this paper, we dispense with the assumption that voters perfectly know their political preferences. Instead, voters can flexibly learn about their ideal points at a cost and do so for the purpose of expressing their political opinion in two-party elections. Voters’ choice set shapes the revealed ideology in the population through learning about ideal points. Because voters’ choice set is constrained to the two party platforms in policy space, voters are not motivated to learn in a

multidimensional or continuous way about their ideal points. As a result, revealed ideology displays issue alignment and polarization. A reduction in the information cost raises party and voter polarization and can harm voters. Parties can adopt positions more extreme than their own ideal points. These results highlight the importance of understanding political behavior as a function not only of preferences or identity, but of the institutional and informational constraints within which decisions are made.

This paper opens several avenues for future research. One important question is what happens when voters learn jointly about their ideal points and party platforms. Such joint learning could explain the correlation between voters' perceptions of candidate positions and their own ideologies (Hare, Armstrong, Bakker, Carroll, and Poole 2015). The model can also be used as a building block to study other questions in political economy, especially in multidimensional policy spaces. Standard models do not allow for closed-form solutions and are usually restricted to one-dimensional policy spaces. Our model allows for a multidimensional policy space and gives closed-form solutions for party positions, voter ideology, and vote shares in a large parameter region.

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A Appendix: Voter Learning

Voter Learning Problem The instrumental utility of τ , that is the voter objective neglecting the information cost, can be rewritten as follows:

$$\begin{aligned}
& \mathbb{E}_\tau \left[\max \left\{ \mathbb{E}_\pi [-(\theta - x_a)^\top A(\theta - x_a)], \mathbb{E}_\pi [-(\theta - x_b)^\top A(\theta - x_b)] \right\} \right] \\
&= \mathbb{E}_\tau \left[\max \left\{ -x_a^\top A x_a + 2x_a^\top A \mathbb{E}_\pi[\theta], -x_b^\top A x_b + 2x_b^\top A \mathbb{E}_\pi[\theta] \right\} - \mathbb{E}_\pi[\theta^\top A \theta] \right] \\
&= \mathbb{E}_\tau \left[\max \left\{ -x_a^\top A x_a + 2x_a^\top A \mathbb{E}_\pi[\theta], -x_b^\top A x_b + 2x_b^\top A \mathbb{E}_\pi[\theta] \right\} \right] - \mathbb{E}_\mu[\theta^\top A \theta] \\
&= \mathbb{E}_\tau \left[\left| (x_b - x_a)^\top A \left(\mathbb{E}_\pi[\theta] - \frac{x_a + x_b}{2} \right) \right| \right] + C.
\end{aligned} \tag{8}$$

where

$$C = \mathbb{E}_\mu \left[(x_a + x_b)^\top A \theta - \theta^\top A \theta \right] - \frac{1}{2} \left(x_a^\top A x_a + x_b^\top A x_b \right)$$

does not depend on τ and can hence be ignored.

By the law of total variance and the fact that the average posterior mean equals the prior mean, the information cost can be rewritten as

$$c(\tau) = \text{Var}_\tau(\mathbb{E}_\pi[\theta]) = \mathbb{E}_\tau [\|\mathbb{E}_\pi[\theta]\|^2].$$

Together, the voter learning problem reads

$$\max_{\substack{\tau \in \Delta(\Delta(\mathbb{R}^n)) \\ \mathbb{E}_\tau[\pi] = \mu}} \mathbb{E}_\tau \left[\left| (x_b - x_a)^\top A \left(\mathbb{E}_\pi[\theta] - \frac{x_a + x_b}{2} \right) \right| - \kappa \|\mathbb{E}_\pi[\theta]\|^2 \right]. \tag{9}$$

The objective depends on τ only through the induced distribution of the posterior mean $\mathbb{E}_\pi[\theta]$. This distribution is well defined for every feasible τ : applying the constraint $\mathbb{E}_\tau[\pi] = \mu$ to the nonnegative function $\theta \mapsto \|\theta\|$ (extending the equality from bounded to nonnegative functions by monotone convergence) yields $\mathbb{E}_\tau[\mathbb{E}_\pi[\|\theta\|]] = \mathbb{E}_\mu[\|\theta\|] < \infty$, so τ -almost every posterior has a finite mean. As is known, a distribution $\rho \in \Delta(\mathbb{R}^n)$ over posterior means is induced by some feasible τ if and only if $\rho \preceq_{\text{MPS}} \mu$. The voter learning problem (9) is therefore equivalent to (4).

A.1 Theorem 1

Proof. We first introduce some terminology. For $\rho \in \Delta(\mathbb{R}^n)$, write

$$b(\rho) = \mathbb{E}_{\theta \sim \rho} \left[\left| (A(x_b - x_a))^\top \left(\theta - \frac{x_a + x_b}{2} \right) \right| \right] \tag{10}$$

for the instrumental value and

$$c(\rho) = \mathbb{E}_{\theta \sim \rho} [\theta^\top \theta]$$

for the cost of ρ , so that the voter's objective (4) equals $b(\rho) - \kappa c(\rho)$. We say ρ is feasible if $\rho \preceq_{\text{MPS}} \mu$. Define

$$v := A(x_b - x_a), \quad L := \text{span}(\{v\}).$$

L is the line through the prior mean with direction $A(x_b - x_a)$.

Fix a feasible ρ . We construct a feasible ρ^* that is supported on L , induces at most two posterior means, and satisfies $b(\rho^*) = b(\rho)$ and $c(\rho^*) \leq c(\rho)$, where the inequality is strict whenever ρ does not place mass 1 on L or induces more than two posterior means. This proves the theorem. We construct ρ^* in four steps.

Step 1: Reflection. Let ref denote the Euclidean reflection around L ,

$$\text{ref}(\theta) := 2 \frac{v^\top \theta}{v^\top v} v - \theta,$$

and denote the induced pushforward on $\Delta(\mathbb{R}^n)$ also by ref . The distribution $\text{ref}(\rho)$ is feasible: since ref is linear, $\rho \preceq_{\text{MPS}} \mu$ implies $\text{ref}(\rho) \preceq_{\text{MPS}} \text{ref}(\mu)$, and $\text{ref}(\mu) = \mu$ because ref is orthogonal and μ is spherical. Moreover, $b(\text{ref}(\rho)) = b(\rho)$, because the instrumental value (10) depends on θ only through $v^\top \theta$ and $v^\top \text{ref}(\theta) = v^\top \theta$. Finally, $c(\text{ref}(\rho)) = c(\rho)$, because ref preserves $\theta^\top \theta$. Together, the voter is indifferent between ρ and $\text{ref}(\rho)$.

Step 2: Randomization. Consider the mixture

$$\rho^m := \frac{1}{2}\rho + \frac{1}{2}\text{ref}(\rho).$$

The mixture is feasible because the set of mean-preserving contractions of μ is convex. Equivalently, ρ^m is induced by the voter randomizing between the two distributions over posteriors that induce ρ and $\text{ref}(\rho)$. Since b and c are linear in ρ , Step 1 implies $b(\rho^m) = b(\rho)$ and $c(\rho^m) = c(\rho)$. Thus the voter is indifferent between ρ and ρ^m . By construction, ρ^m is symmetric around L in the sense that $\text{ref}(\rho^m) = \rho^m$, which we use in the next step.

Step 3: Projection. Let

$$\text{Proj}_L(\theta) := \frac{v^\top \theta}{v^\top v} v$$

denote the orthogonal projection onto L , extended to distributions via the pushforward. We claim

$$\rho^p := \text{Proj}_L(\rho^m)$$

is a mean-preserving contraction of ρ^m . Let $Y \sim \rho^m$ and $X := \text{Proj}_L(Y)$. Because $\text{ref}(\rho^m) = \rho^m$ and $\text{Proj}_L \circ \text{ref} = \text{Proj}_L$, the conditional distribution of Y given X is invariant under ref , so $\mathbb{E}[Y | X]$ is a fixed point of ref and hence lies on L . Since $\text{Proj}_L(\mathbb{E}[Y | X]) = \mathbb{E}[\text{Proj}_L(Y) | X] = X$, it follows that $\mathbb{E}[Y | X] = X$. Thus ρ^p is a mean-preserving contraction of ρ^m and hence feasible.

The instrumental value is unchanged, $b(\rho^p) = b(\rho^m)$, because $v^\top \text{Proj}_L(\theta) = v^\top \theta$. The cost weakly decreases: pointwise, $\text{Proj}_L(\theta)^\top \text{Proj}_L(\theta) \leq \theta^\top \theta$, with strict inequality for $\theta \notin L$. Hence, if ρ (and therefore ρ^m) does not place mass 1 on L , then $c(\rho^p) < c(\rho^m) = c(\rho)$.

Step 4: Binarization. Finally, we replace ρ^p by the at most binary distribution given by its conditional means on the two half-spaces in which voting for a and b , respectively, is optimal. This corresponds to a garbling based on the action recommendation, which preserves the instrumental

value and weakly reduces the cost. Partition the state space into the half-spaces

$$R_a := \{\theta \in \mathbb{R}^n \mid v^\top(\theta - \frac{x_a+x_b}{2}) \leq 0\}, \quad R_b := \{\theta \in \mathbb{R}^n \mid v^\top(\theta - \frac{x_a+x_b}{2}) > 0\}.$$

Let ρ^* denote the (at most binary) distribution placing mass $\rho^p(R_j)$ on the conditional mean $\mathbb{E}_{\rho^p}[\theta \mid R_j]$ for each half-space with positive probability. The distribution ρ^* is a mean-preserving contraction of ρ^p and hence feasible, ρ^* induces at most two posterior means, and ρ^* places mass 1 on L because ρ^p does. The instrumental value is preserved, $b(\rho^*) = b(\rho^p)$, because on each half-space, $\theta \mapsto |v^\top(\theta - \frac{x_a+x_b}{2})|$ is affine. The cost weakly decreases, $c(\rho^*) \leq c(\rho^p)$, because ρ^* is a mean-preserving contraction of ρ^p and $\theta^\top \theta$ is convex in θ . If ρ^p has support greater than two, then the conditional distribution on at least one cell is non-degenerate, so strict Jensen's inequality for the strictly convex map $\theta \mapsto \theta^\top \theta$ yields $c(\rho^*) < c(\rho^p)$.

Conclusion. The distribution ρ^* is feasible, supported on L with at most two support points, and satisfies $b(\rho^*) = b(\rho)$ and $c(\rho^*) \leq c(\rho)$. If ρ does not place mass 1 on L , the inequality is strict by Step 3. If ρ places mass 1 on L but induces more than two posterior means, then $\rho^p = \rho^m = \rho$, and the inequality is strict by Step 4. Hence any solution to the voter problem induces at most two posterior means and places mass 1 on L . $\square$

A.2 Lemma 1

Proof. Let $M := \{\rho \in \Delta(\mathbb{R}^n) \mid \rho \preceq_{\text{MPS}} \mu\}$ denote the feasible set of (4). By Strassen's theorem, the MPS order coincides with the convex order: $\rho \preceq_{\text{MPS}} \mu$ if and only if $\int \phi d\rho \leq \int \phi d\mu$ for every convex $\phi: \mathbb{R}^n \rightarrow \mathbb{R}$. In particular, M is convex, every $\rho \in M$ has mean $\mathbb{E}_\rho[\theta] = \mathbb{E}_\mu[\theta] = 0$, and $\mathbb{E}_\rho[\|\theta\|^k] \leq \mathbb{E}_\mu[\|\theta\|^k] < \infty$ for every $k \geq 1$, where all moments of μ are finite by log-concavity.

M is compact: Because $\mathbb{R}^n$ is Polish, $\Delta(\mathbb{R}^n)$ equipped with the weak topology is Polish as well, so Prokhorov's theorem applies: tightness is equivalent to relative compactness. M is tight because Markov's inequality and convexity of $\phi(\theta) = \|\theta\|$ give $\rho(\|\theta\| > R) \leq \mathbb{E}_\rho[\|\theta\|]/R \leq \mathbb{E}_\mu[\|\theta\|]/R$ uniformly over $\rho \in M$. M is closed: let $\rho_k \in M$ converge weakly to ρ . The uniform second-moment bound makes first moments uniformly integrable, so $\mathbb{E}_\rho[\theta] = \lim_k \mathbb{E}_{\rho_k}[\theta] = 0$. Now let ϕ be convex and let a be an affine function with $a \leq \phi$. The integral of a under any mean-zero distribution equals its intercept, so the Portmanteau theorem applied to the non-negative continuous function $\phi - a$ yields

$$\int \phi d\rho \leq \liminf_k \int \phi d\rho_k \leq \int \phi d\mu.$$

Hence $\rho \in M$, and M is compact.

Existence: The integrand of (4) is continuous and bounded above by some $C > 0$, because its quadratic term dominates its piecewise linear term for large $\|\theta\|$. The objective is therefore weakly upper semicontinuous by the Portmanteau. An upper semicontinuous function on the compact set M attains its maximum.

Uniqueness: As in the proof of Theorem 1, partition the state space into the half-spaces

$$R_a = \{\theta \in \mathbb{R}^n \mid v^\top(\theta - \frac{x_a+x_b}{2}) \leq 0\}, \quad R_b = \{\theta \in \mathbb{R}^n \mid v^\top(\theta - \frac{x_a+x_b}{2}) > 0\}.$$

The binarization argument in Step 4 of the proof of Theorem 1 applies verbatim to any feasible ρ : unless ρ equals its binarization, the binarization is feasible, attains the same instrumental value, and has strictly smaller cost. Hence any optimal ρ is supported on at most one point in each half-space. Now let ρ_1 and ρ_2 be optimal. Because M is convex and the objective is linear in ρ , the mixture $\rho^m := \frac{1}{2}\rho_1 + \frac{1}{2}\rho_2$ is optimal, and its support is $\text{supp } \rho_1 \cup \text{supp } \rho_2$. Since ρ^m is supported on at most one point per half-space, ρ_1 and ρ_2 are supported on a common set $\{\theta_a, \theta_b\}$, where θ_j denotes the point in R_j , if any. If this set is a singleton $\{\theta\}$, then $\rho_1 = \rho_2 = \delta_\theta$. Otherwise, each ρ_i places some weight w_i on θ_a with $w_i\theta_a + (1-w_i)\theta_b = 0$, because every $\rho \in M$ has mean zero; since $\theta_a \neq \theta_b$, this equation has at most one solution, so $w_1 = w_2$ and $\rho_1 = \rho_2$. $\square$

A.3 Additional Results on Voter Learning

The previous results use only certain properties of the information cost. This section uses the posterior-variance information cost to characterize the solution to the voter learning problem and the resulting vote shares. We begin with the one-dimensional case $n = 1$, in which party platforms are scalars $x_a, x_b \in \mathbb{R}$ and the voter prior $\mu \in \Delta(\mathbb{R})$ has a log-concave density f_μ that is symmetric around 0. At the end of the section we turn to the general n -dimensional case and show how it reduces to the one-dimensional one with transformed parameters.

In one dimension, the posterior-mean formulation of the voter learning problem is

$$\begin{aligned} \max_{\rho \in \Delta(\mathbb{R})} \mathbb{E}_{\theta \sim \rho} \left[\left| (x_b - x_a) \left(\theta - \frac{x_a+x_b}{2} \right) \right| - \kappa \theta^2 \right] \\ \text{s.t. } \rho \preceq_{\text{MPS}} \mu. \end{aligned} \tag{11}$$

The following definition will be useful below.

Definition 1. For $\theta_a, \theta_b \in \mathbb{R}$, we say $\{\theta_a, \theta_b\}$ is **feasible** if there exists a distribution ρ on $\{\theta_a, \theta_b\}$ that is a mean-preserving contraction of the prior μ .

If ρ is a mean-preserving contraction of μ , then ρ has the same expectation as μ , which is 0. So if $0 \notin [\theta_a, \theta_b]$, then $\{\theta_a, \theta_b\}$ is not feasible. If instead $0 \in [\theta_a, \theta_b]$, then there is a unique distribution ρ over $\{\theta_a, \theta_b\}$ with expectation 0. Feasibility therefore reduces to whether this distribution ρ is a mean-preserving contraction of the prior μ .

We fix $x_a < x_b$ in the following. The case $x_a > x_b$ is analogous and the case $x_a = x_b$ is trivial since it induces no voter learning. The definition (6) of the posterior means from the multidimensional case simplifies in one dimension to

$$\begin{aligned} \theta_a &:= \frac{x_a+x_b}{2} - \frac{x_b-x_a}{2\kappa}, \\ \theta_b &:= \frac{x_a+x_b}{2} + \frac{x_b-x_a}{2\kappa}. \end{aligned}$$

By Lemma 1 a solution to the voter learning problem (11) exists and is unique. The following proposition characterizes it.

Proposition 2 (Revealed Ideal Points). *The solution ρ to (11) is characterized by:*

- (1) *If $\{\theta_a, \theta_b\}$ is feasible, ρ is the mean-zero distribution supported on $\{\theta_a, \theta_b\}$.*
- (2) *If $0 \notin [\theta_a, \theta_b]$, $\rho = \delta_0$.*
- (3) *If $\{\theta_a, \theta_b\}$ is not feasible and $0 \in [\theta_a, \theta_b]$, the optimal ρ is induced by a threshold signal structure $s(\theta) = \mathbb{1}(\theta > t)$, where $t \in \mathbb{R} \cup \{-\infty, \infty\}$ is more extreme than the party midpoint, that is, either $0 < \frac{x_a + x_b}{2} < t$, $0 = \frac{x_a + x_b}{2} = t$, or $t < \frac{x_a + x_b}{2} < 0$. The induced posterior means are in $[\theta_a, \theta_b]$. As κ decreases, the threshold t moves weakly closer to the party midpoint $\frac{x_a + x_b}{2}$.*

When voters acquire information, one posterior mean θ_a is strictly closer to x_a and the other, θ_b , is strictly closer to x_b . The vote shares of parties are the probabilities of posterior means θ_a and θ_b . These probabilities are uniquely pinned down by $\Pr_\rho(\theta_a)\theta_a + \Pr_\rho(\theta_b)\theta_b = 0$. The following result summarizes the implications for parties' vote shares.

Corollary 3 (Vote Shares). *The vote shares induced by the solution ρ to (11) are characterized by:*

- (1) *If $\{\theta_a, \theta_b\}$ is feasible,*

$$P_a(x_a, x_b) = \frac{1}{2} + \frac{\kappa x_b + x_a}{2 x_b - x_a}.$$
- (2) *If $0 \notin (\theta_a, \theta_b)$, the party whose position is closer to 0 obtains all votes.*
- (3) *If $\{\theta_a, \theta_b\}$ is not feasible and $0 \in (\theta_a, \theta_b)$, $P_a(x_a, x_b) = F_\mu(t)$, where t is as in Proposition 2. If party a is more extreme, that is, $|x_a| > |x_b|$, then $P_a(x_a, x_b) < \frac{1}{2} + \frac{\kappa x_b + x_a}{2 x_b - x_a}$.*

As a further corollary, we obtain the bias toward the moderate party in one dimension.

Corollary 4 (Bias toward Moderate Party). *If $|x_a| < |x_b|$, then party a 's vote share $P_a(x_a, x_b)$ weakly exceeds the share of voters whose true ideal point is closer to x_a than to x_b . The vote share $P_a(x_a, x_b)$ weakly increases if we raise the information cost parameter κ , and weakly decreases if we raise $x_b - x_a$ while holding the midpoint $\frac{x_a + x_b}{2}$ fixed.*

Feasibility: We now give a sufficient condition for feasibility of $\{\theta_a, \theta_b\}$ based on the following lemma due to Jewitt (2004). Define $\Delta(t) := \mathbb{E}[\theta \mid \theta > t] - \mathbb{E}[\theta \mid \theta < t]$.

Lemma 2 (Jewitt's Lemma). *The map $\Delta(t)$ is weakly decreasing in t for $t < 0$ and weakly increasing in t for $t > 0$.*

Proof. Jewitt (2004) shows that $\Delta(t)$ is quasiconvex when the density of θ is quasiconcave. A log-concave density is quasiconcave, so $\Delta(t)$ is quasiconvex. Because the density of θ is symmetric around 0, $\Delta(t)$ is symmetric around 0, so $\Delta(t)$ is weakly decreasing for $t < 0$ and weakly increasing for $t > 0$. $\square$

Define $\Delta := \Delta(0) = \mathbb{E}[\theta | \theta > 0] - \mathbb{E}[\theta | \theta < 0]$. By symmetry of μ , $\Delta = 2\mathbb{E}[\theta | \theta > 0] = 2\mathbb{E}[|\theta|]$.

Lemma 3. *If $|x_b - x_a| \leq \kappa\Delta$ and $\theta_a \leq 0 \leq \theta_b$, then $\{\theta_a, \theta_b\}$ is feasible.*

Proof. The posterior means θ_a, θ_b satisfy $|\theta_b - \theta_a| = |x_b - x_a|/\kappa \leq \Delta$. Since $\theta_a \leq 0 \leq \theta_b$ and $\theta_b - \theta_a \leq \Delta$, at least one of θ_a, θ_b has absolute value at most $\Delta/2$. Suppose $|\theta_a| \leq \Delta/2$, the case $|\theta_b| \leq \Delta/2$ being analogous.

A threshold signal with threshold $t = 0$ induces the posterior means $\mathbb{E}[\theta | \theta < t] = -\Delta/2$ and $\mathbb{E}[\theta | \theta > t] = \Delta/2$. As t increases to ∞ , $\mathbb{E}[\theta | \theta < t]$ converges to $\mathbb{E}[\theta] = 0$. The function $\mathbb{E}[\theta | \theta < t]$ is continuous in t because θ has a density, so there is a $t' \geq 0$ with $\mathbb{E}[\theta | \theta < t'] = \theta_a$. By Jewitt's Lemma and $\theta_b - \theta_a \leq \Delta$, $\mathbb{E}[\theta | \theta > t'] \geq \theta_b$. The mean-zero distribution on $\{\theta_a, \theta_b\}$ is a mean-preserving contraction of the mean-zero distribution on $\{\mathbb{E}[\theta | \theta < t'], \mathbb{E}[\theta | \theta > t']\}$, since $\theta_a \leq 0 \leq \theta_b \leq \mathbb{E}[\theta | \theta > t']$. Hence $\{\theta_a, \theta_b\}$ is feasible. $\square$

Dimension Reduction: We now turn to the general n -dimensional case and show that it reduces to the one-dimensional problem solved above. From here on x_a, x_b , and μ are again the general primitives, $x_a, x_b \in \mathbb{R}^n$ and $\mu \in \Delta(\mathbb{R}^n)$. Assume $x_a \neq x_b$, so that the direction $A(x_b - x_a)$ is well defined. The voter problem is

$$\begin{aligned} \max_{\rho \in \Delta(\mathbb{R}^n)} \mathbb{E}_{\theta \sim \rho} \left[\left| (x_b - x_a)^\top A\left(\theta - \frac{x_a + x_b}{2}\right) \right| - \kappa \theta^\top \theta \right] \\ \text{s.t. } \rho \preceq_{\text{MPS}} \mu. \end{aligned} \quad (\text{Pn})$$

Let proj_v denote the scalar projection on $v \in \mathbb{R}^n$,

$$\begin{aligned} \text{proj}_v : \mathbb{R}^n &\rightarrow \mathbb{R} \\ x &\mapsto \frac{v^\top x}{\|v\|}, \end{aligned} \quad (12)$$

so $\text{proj}_{A(x_b - x_a)}$ is the scalar projection onto the line through the origin with direction $A(x_b - x_a)$, extended to measures by the pushforward. Define

$$\hat{x}_a = \text{proj}_{A(x_b - x_a)}(x_a), \quad \hat{x}_b = \text{proj}_{A(x_b - x_a)}(x_b) \quad (13)$$

$$\hat{\kappa} = \frac{(x_b - x_a)^\top A(x_b - x_a)}{\|A(x_b - x_a)\|^2} \kappa, \quad (14)$$

$$\hat{\mu} = \text{proj}_{A(x_b - x_a)}(\mu). \quad (15)$$

The associated one-dimensional problem is

$$\begin{aligned} \max_{\hat{\rho} \in \Delta(\mathbb{R})} \mathbb{E}_{\hat{\theta} \sim \hat{\rho}} \left[\left| (\hat{x}_b - \hat{x}_a)(\hat{\theta} - \frac{\hat{x}_a + \hat{x}_b}{2}) \right| - \hat{\kappa} \hat{\theta}^2 \right] \\ \text{s.t. } \hat{\rho} \preceq_{\text{MPS}} \hat{\mu}. \end{aligned} \quad (\text{P1})$$

By Theorem 1, revealed ideal points are supported on the line through the origin with direction $A(x_b - x_a)$. Along that line the two problems coincide up to a positive scaling of the objective, as

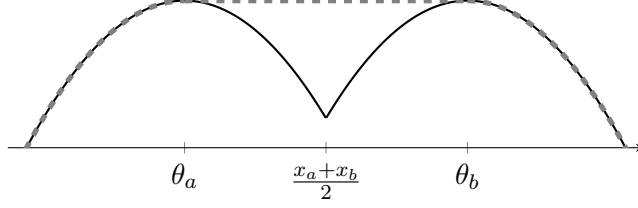


Figure 6: The value function $v(\theta)$ and its concave envelope (dashed).

the following lemma records.

Lemma 4. *A distribution $\rho \in \Delta(\mathbb{R}^n)$ solves (Pn) if and only if ρ is supported on the line through the origin with direction $A(x_b - x_a)$ and $\hat{\rho} := \text{proj}_{A(x_b - x_a)}(\rho)$ solves (P1).*

If μ is spherical and log-concave, then $\hat{\mu}$ is symmetric around 0 and log-concave, the latter because log-concavity is preserved by marginalization. Thus, problem (P1) has exactly the form solved above, with cost $\hat{\kappa}$, platforms $\hat{x}_a, \hat{x}_b$, and prior $\hat{\mu}$. Specializing Proposition 2 and Corollaries 3 and 4 to (P1) and projecting back therefore characterizes revealed ideal points and vote shares in the general case, and gives the bias result of Proposition 1.

A.3.1 Proposition 2

Proof. Recall that we assume $x_a < x_b$. We rewrite the voter learning problem using the Rothschild-Stiglitz approach (Gentzkow and Kamenica 2016):

$$\max_{\rho \in \Delta(\mathbb{R})} \mathbb{E}_{\theta \sim \rho} \left[\overbrace{\left| (x_b - x_a) \left(\theta - \frac{x_a + x_b}{2} \right) \right| - \kappa \left(\theta - \frac{x_a + x_b}{2} \right)^2}^{v(\theta) :=} \right] \quad (16)$$

$$\text{s.t.} \quad \mathbb{E}_{\theta \sim \rho}[\theta] = 0, \quad (17)$$

$$\int_{-\infty}^t F_{\rho}(\theta) d\theta \leq \int_{-\infty}^t F_{\mu}(\theta) d\theta \quad \forall t \in \mathbb{R}. \quad (18)$$

This reformulation replaces the mean-preserving contraction constraint $\rho \preceq_{\text{MPS}} \mu$ by the mean constraint (17) and the majorization constraints (18). By (17), adding an affine function to the integrand of (11) amounts to adding a constant, so the optimum is unchanged.

We first solve the relaxed problem that drops the majorization constraints (18). This is a one-dimensional persuasion problem, solved by concavification of the value function v in posterior-mean space (Aumann and Maschler 1995; Kamenica and Gentzkow 2011). The value function v , shown with its concave envelope in Figure 6, has maxima at $\theta_a = \frac{x_a + x_b}{2} + \frac{x_b - x_a}{2\kappa}$ and $\theta_b = \frac{x_a + x_b}{2} - \frac{x_b - x_a}{2\kappa}$. If $0 \notin [\theta_a, \theta_b]$, the solution is δ_0 . If $0 \in [\theta_a, \theta_b]$, the solution is the mean-zero distribution ρ^* supported on $\{\theta_a, \theta_b\}$.

By Lemma 1, the original problem (16) has a unique solution. If the solution to the relaxed problem satisfies the majorization constraints, then it solves the original problem. The point mass δ_0 is always feasible, and ρ^* satisfies $\rho^* \preceq_{\text{MPS}} \mu$ exactly when $\{\theta_a, \theta_b\}$ is feasible. This covers the first two cases of the proposition.

It remains to treat the case $0 \in [\theta_a, \theta_b]$ when $\{\theta_a, \theta_b\}$ is not feasible, for which we show that the optimal ρ is induced by a threshold signal $s(\theta) = \mathbf{1}(\theta > t)$ with $t \in \mathbb{R} \cup \{-\infty, \infty\}$ with posterior means $\mathbb{E}_\mu[\theta | \theta \leq t]$ and $\mathbb{E}_\mu[\theta | \theta > t]$. By Theorem 1, the solution ρ to (16) is supported on at most two points. If the support is a single point, then $\rho = \delta_0$, which is induced by the threshold signal with threshold $t = \pm\infty$.

Suppose instead ρ is supported on two points $\{y, z\}$ with $y < 0 < z$, implying $\rho(y) = \frac{z}{z-y}$. The resulting value of the voter problem (11) is $V(y, z) = \frac{z}{z-y}v(y) + \frac{-y}{z-y}v(z)$.

First, $\theta_a \leq y < z \leq \theta_b$. If a support point lay strictly outside $[\theta_a, \theta_b]$, say $z > \theta_b$, then by $0 < \theta_b < z$ there is a mean-zero distribution ρ' on $\{y, \theta_b\}$ that is a mean-preserving contraction of ρ , hence feasible. By $\rho'(\theta_b) > \rho(z)$ and $v(\theta_b) \geq \max\{v(y), v(z)\}$, we obtain $V(y, \theta_b) > V(y, z)$, contradicting optimality.

Second, the majorization constraint binds at some $t \in [y, z]$. Write

$$G_{y,z}(t) := \int_{-\infty}^t (F_\mu - F_\rho) d\theta = \int_{-\infty}^t F_\mu(\theta) d\theta - \left(\frac{z}{z-y}(t-y)^+ + \frac{-y}{z-y}(t-z)^+ \right),$$

so the majorization constraint (18) reads $G_{y,z}(t) \geq 0$ for all $t \in \mathbb{R}$. Toward a contradiction, suppose $G_{y,z}(t) > 0$ for all $t \in [y, z]$. We show the mean-zero distribution ρ' on $\{ky, kz\}$ with $k > 1$ close enough to 1 is an improvement. Scaling preserves the probabilities, so $\rho'(ky) = \rho(y)$ and $\rho'(kz) = \rho(z)$. Without loss suppose $y \geq \theta_a$ and $z < \theta_b$. Optimality requires $y < \frac{x_a+x_b}{2} < z$. Then $v'(y) \leq 0$ and $v'(z) > 0$, so $\frac{d}{dk}V(ky, kz) = y\rho(y)v'(y) + z\rho(z)v'(z) > 0$ at $k = 1$, so for k close enough to 1, $V(ky, kz) > V(y, z)$. Finally, ρ' is feasible for k close enough to 1. The map $(y, z, t) \mapsto G_{y,z}(t)$ is jointly continuous, so by the maximum theorem $\min_{t \in [y, z]} G_{y,z}(t) > 0$ and $\min_{t \in [y', z']} G_{y',z'}(t) > 0$ for (y', z') close enough to (y, z) . Outside $[y', z']$ the majorization constraint holds automatically: since $F_{\rho'} = 0$ below y' , the integrand $F_\mu - F_{\rho'}$ is nonnegative there, so $G_{y',z'}$ increases from $G_{y',z'}(-\infty) = 0$ and stays nonnegative; since $F_{\rho'} = 1$ above z' , the integrand is nonpositive there, so $G_{y',z'}$ decreases to $G_{y',z'}(+\infty) = 0$, the latter by the mean-zero property, and again stays nonnegative. Hence ρ' satisfies (18) for all t , contradicting optimality. Therefore $G_{y,z}(t) = 0$ for some $t \in [y, z]$.

A binary distribution with support $\{y, z\}$ whose majorization constraint binds at $t \in [y, z]$ is induced by the threshold signal with threshold t , with posterior means $y = \mathbb{E}_\mu[\theta | \theta \leq t]$ and $z = \mathbb{E}_\mu[\theta | \theta > t]$.

The optimal threshold t is strictly more extreme than $\frac{x_a+x_b}{2}$ (that is, t is strictly further from 0 than is $\frac{x_a+x_b}{2}$), as we now show. One can easily verify that the instrumental value of information $b(t)$ increases as t moves toward $\frac{x_a+x_b}{2}$:

$$\begin{aligned} b(t) &:= \int_{-\infty}^t -(x_a - \theta)^2 f_\mu(\theta) d\theta + \int_t^\infty -(x_b - \theta)^2 f_\mu(\theta) d\theta \\ \Rightarrow b'(t) &= (-(x_a - t)^2 + (x_b - t)^2) f_\mu(t) = 2(t - \frac{x_a+x_b}{2})(x_b - x_a) f_\mu(t). \end{aligned}$$

In particular, $b'(\frac{x_a+x_b}{2}) = 0$. The information cost decreases as t becomes more extreme, provided

the prior is log-concave. The information cost is

$$\begin{aligned}
c(t) &:= \text{Var}[\theta] - (F_\mu(t) \text{Var}[\theta|\theta \leq t] + (1 - F_\mu(t)) \text{Var}[\theta|\theta > t]) \\
&= \text{Var}[\theta] - \left(F_\mu(t) \min_E \left\{ \frac{\int_{-\infty}^t (\theta - E)^2 dF_\mu(\theta)}{F_\mu(t)} \right\} + (1 - F_\mu(t)) \min_E \left\{ \frac{\int_t^\infty (\theta - E)^2 dF_\mu(\theta)}{1 - F_\mu(t)} \right\} \right) \\
&= \text{Var}[\theta] - \left(\min_E \left\{ \int_{-\infty}^t (\theta - E)^2 f_\mu(\theta) d\theta \right\} + \min_E \left\{ \int_t^\infty (\theta - E)^2 f_\mu(\theta) d\theta \right\} \right)
\end{aligned}$$

By the envelope theorem,

$$\begin{aligned}
c'(t) &= f_\mu(t)(t - \mathbb{E}[\theta|\theta > t])^2 - f_\mu(t)(t - \mathbb{E}[\theta|\theta \leq t])^2 \\
&= f_\mu(t)(\mathbb{E}[\theta - t|\theta > t]^2 - \mathbb{E}[t - \theta|\theta \leq t]^2).
\end{aligned}$$

By Theorem 1.C.52 in Shaked and Shanthikumar (2007), $\mathbb{E}[\theta - t|\theta > t]$ is decreasing in t when f_μ is log-concave. By symmetry of f_μ around 0, $\mathbb{E}[t - \theta|\theta \leq t] = \mathbb{E}[\theta - (-t)|\theta > -t]$. Hence, for $t > 0$,

$$0 < \mathbb{E}[\theta - t|\theta > t] < \mathbb{E}[\theta|\theta > 0] < \mathbb{E}[\theta - (-t)|\theta > -t] = \mathbb{E}[t - \theta|\theta \leq t],$$

so $c'(t) < 0$ for $t > 0$.

For $\frac{x_a + x_b}{2} > 0$, any threshold $t < 0$ is dominated by its reflection $-t > 0$, which has the same information cost and strictly higher instrumental value. For $0 \leq t < \frac{x_a + x_b}{2}$, we have $b'(t) > 0$ and $c'(t) \leq 0$, so $t = \frac{x_a + x_b}{2}$ dominates every $t < \frac{x_a + x_b}{2}$. Finally, $b'(\frac{x_a + x_b}{2}) = 0$ and $c'(\frac{x_a + x_b}{2}) < 0$ give $\frac{d}{dt}[b(t) - \kappa c(t)]|_{t=\frac{x_a + x_b}{2}} > 0$, so every optimal t satisfies $t > \frac{x_a + x_b}{2}$. The optimal distribution over posteriors is unique. By log-concavity of the prior density, μ has connected support, so the optimal threshold t is unique, unless $t = +\infty$ or $t = -\infty$ is optimal, in which case every threshold beyond the prior support on the corresponding side is also optimal.

Restrict t to $[\frac{x_a + x_b}{2}, \infty]$, where $t = \infty$ denotes always voting for party a . Since $c'(t) < 0$, the objective $b(t) - \kappa c(t)$ is supermodular in (κ, t) , so the optimal t weakly increases in κ . $\square$

A.3.2 Corollary 3

Proof. Denote $P_a := P_a(x_a, x_b)$.

Case (1): By ρ having mean zero, the vote share P_a satisfies

$$P_a \left(\frac{x_a + x_b}{2} - \frac{x_b - x_a}{2\kappa} \right) + (1 - P_a) \left(\frac{x_a + x_b}{2} + \frac{x_b - x_a}{2\kappa} \right) = 0 \Rightarrow P_a = \frac{1}{2} + \frac{\kappa}{2} \frac{x_a + x_b}{x_b - x_a}.$$

Case (2) is immediate.

Case (3): By case (3) of Proposition 2, $P_a = F_\mu(t)$. It remains to show that, if $\{\theta_a, \theta_b\}$ is not feasible, $0 \in [\theta_a, \theta_b]$, and $|x_a| > |x_b|$, then

$$P_a < \tilde{P}_a := \frac{1}{2} + \frac{\kappa}{2} \frac{x_a + x_b}{x_b - x_a}.$$

Since $x_a < x_b$ and $|x_a| > |x_b|$, we have $\frac{x_a+x_b}{2} < 0$, and hence $\tilde{P}_a < 1/2$. By $\theta_a < 0 < \theta_b$, $\tilde{P}_a = \theta_b/(\theta_b - \theta_a) \in (0, 1)$.

If the voter acquires no information, the posterior mean is 0, so all voters choose party b , which is closer to 0. Hence $P_a = 0 < \tilde{P}_a$. Suppose, therefore, that the voter acquires two posterior means. By Proposition 2, the optimal signal is a threshold signal, and an optimal threshold t induces posterior means satisfying

$$\theta_a \leq \mathbb{E}[\theta \mid \theta < t] < \frac{x_a+x_b}{2} < \mathbb{E}[\theta \mid \theta \geq t] \leq \theta_b. \quad (19)$$

Let t' be defined by $F_\mu(t') = \tilde{P}_a$. Since $\tilde{P}_a < 1/2$ and μ is symmetric, $t' < 0$. By Bayes plausibility,

$$0 = \tilde{P}_a \mathbb{E}[\theta \mid \theta < t'] + (1 - \tilde{P}_a) \mathbb{E}[\theta \mid \theta \geq t'] = \tilde{P}_a \theta_a + (1 - \tilde{P}_a) \theta_b. \quad (20)$$

Hence there exists $\alpha > 0$ such that

$$(\mathbb{E}[\theta \mid \theta < t'], \mathbb{E}[\theta \mid \theta \geq t']) = (\alpha \theta_a, \alpha \theta_b).$$

Since $\{\theta_a, \theta_b\}$ is not feasible, $\alpha < 1$: otherwise $\{\theta_a, \theta_b\}$ would be a mean-preserving contraction of the feasible threshold distribution induced by t' .

We show that no optimal threshold satisfies $t \geq t'$. If $\mathbb{E}[\theta \mid \theta < t'] \geq \frac{x_a+x_b}{2}$, this is immediate from (19), since $\mathbb{E}[\theta \mid \theta < t]$ is increasing in t . Hence suppose $\mathbb{E}[\theta \mid \theta < t'] < \frac{x_a+x_b}{2}$. By Proposition 2, since $\frac{x_a+x_b}{2} < 0$, an optimal threshold satisfies $t < \frac{x_a+x_b}{2} < 0$. Hence, to rule out $t \geq t'$, it is enough to consider candidate thresholds $t \in [t', 0]$ satisfying (19).

By Proposition 2, since $\frac{x_a+x_b}{2} < 0$, any informative optimal threshold satisfies $t < \frac{x_a+x_b}{2} < 0$. Hence, to rule out $t \geq t'$, it is enough to consider candidate thresholds $t \in [t', 0]$ satisfying (19).

Write

$$L(t) := \mathbb{E}[\theta \mid \theta < t], \quad R(t) := \mathbb{E}[\theta \mid \theta \geq t].$$

The objective of the voter problem is $F_\mu(t)v(L(t)) + (1 - F_\mu(t))v(R(t))$. We show that for any $t \in [t', 0]$ satisfying (19),

$$\begin{aligned} \frac{d}{dt}(F_\mu(t)v(L(t)) + (1 - F_\mu(t))v(R(t))) &= (F_\mu(t)v'(L(t)) + (1 - F_\mu(t))v'(R(t)))R'(t) \\ &\quad + F_\mu(t)v'(L(t))(L'(t) - R'(t)) \\ &\quad + f_\mu(t)(v(L(t)) - v(R(t))) < 0, \end{aligned}$$

hence t is suboptimal. We go through the three terms in turn.

The first term $(F_\mu(t)v'(L(t)) + (1 - F_\mu(t))v'(R(t)))R'(t)$ is weakly negative. By (19),

$$v'(L(t)) = -2\kappa(L(t) - \theta_a) \leq 0, \quad v'(R(t)) = 2\kappa(\theta_b - R(t)) \geq 0. \quad (21)$$

If $v'(R(t)) = 0$, we are done. If $v'(R(t)) > 0$, then for any $t \geq t'$ satisfying (19),

$$\frac{-v'(L(t))}{v'(R(t))} = \frac{L(t) - \theta_a}{\theta_b - R(t)} \geq \frac{L(t') - \theta_a}{\theta_b - R(t')} = \frac{\alpha\theta_a - \theta_a}{\theta_b - \alpha\theta_b} = \frac{-\theta_a}{\theta_b} = \frac{1 - F_\mu(t')}{F_\mu(t')}. \quad (22)$$

The inequality uses that both $L(t)$ and $R(t)$ are nondecreasing in t and the last equality uses (20). By $F_\mu(t) \geq F_\mu(t')$ for $t \geq t'$ and (21), the previous display implies

$$F_\mu(t)v'(L(t)) + (1 - F_\mu(t))v'(R(t)) \leq 0. \quad (23)$$

The second term $F_\mu(t)v'(L(t))(L'(t) - R'(t))$ is weakly negative. By (21), $v'(L(t)) < 0$. By Jewitt's Lemma, $\Delta(t) = R(t) - L(t)$ is nonincreasing in t for $t < 0$. Hence $L'(t) - R'(t) \geq 0$.

The third term $f_\mu(t)(v(L(t)) - v(R(t)))$ is strictly negative. Since v is symmetric around $\frac{x_a + x_b}{2}$ and $v(\theta_a) = v(\theta_b)$, it is enough to compare the distances of $\alpha\theta_a$ and $\alpha\theta_b$ from $\frac{x_a + x_b}{2}$. Because $\theta_a < \frac{x_a + x_b}{2} < 0$ and $\alpha < 1$, $|\alpha\theta_b - \frac{x_a + x_b}{2}| < |\alpha\theta_b - \frac{x_a + x_b}{2}|$, so $v(\alpha\theta_a) < v(\alpha\theta_b)$. For $t \geq t'$, $L(t) \geq L(t') = \alpha\theta_a$ and $R(t) \geq R(t') = \alpha\theta_b$. Since v is decreasing on $[\theta_a, \frac{x_a + x_b}{2}]$ and increasing on $[\frac{x_a + x_b}{2}, \theta_b]$, (19) implies $v(L(t)) \leq v(\alpha\theta_a) < v(\alpha\theta_b) \leq v(R(t))$.

Thus the optimal threshold satisfies $t < t'$ and

$$P_a(x_a, x_b) = F_\mu(t) < F_\mu(t') = \tilde{P}_a = \frac{1}{2} + \frac{\kappa}{2} \frac{x_a + x_b}{x_b - x_a}. \quad \square$$

A.3.3 Corollary 4

Proof. We go through the three cases of Corollary 3 in order of increasing κ . We show in each case the vote share is biased toward the moderate party, and the bias weakly increases in κ . Dividing the objective in (11) by $x_b - x_a > 0$ does not change its maximizer and rewrites it as $|\theta - \frac{x_a + x_b}{2}| - \frac{\kappa}{x_b - x_a} \theta^2$, so the solution depends on $x_b - x_a$ and κ only through $\frac{\kappa}{x_b - x_a}$. Raising $x_b - x_a$ while holding $\frac{x_a + x_b}{2}$ fixed lowers this ratio exactly as lowering κ does, so the comparative static in $x_b - x_a$ follows from the one in κ . Throughout, $\frac{x_a + x_b}{2} > 0$ because $x_a < x_b$ and $|x_a| < |x_b|$, so party a is the moderate party and a voter is closer to x_a than to x_b exactly when $\theta < \frac{x_a + x_b}{2}$. The share of voters whose true ideal point is closer to x_a than to x_b is therefore $F_\mu(\frac{x_a + x_b}{2})$.

For small κ we have $0 \in [\theta_a, \theta_b]$, and $\{\theta_a, \theta_b\}$ is not feasible because θ_a and θ_b diverge as $\kappa \rightarrow 0$. By Corollary 3 the solution is induced by a threshold signal with threshold t strictly more extreme than the midpoint, so $\frac{x_a + x_b}{2} < t$ and $P_a(x_a, x_b) = F_\mu(t) \geq F_\mu(\frac{x_a + x_b}{2})$. As κ increases, t moves weakly farther from $\frac{x_a + x_b}{2}$ by Proposition 2, so the vote share $F_\mu(t)$ weakly increases.

While $0 \in [\theta_a, \theta_b]$, raising κ keeps $\{\theta_a, \theta_b\}$ feasible once it is feasible: raising κ moves θ_a, θ_b closer together which corresponds to a mean-preserving contraction. The set of mean-preserving contractions of μ is compact (Lemma 1), so if $\{\theta_a, \theta_b\}$ is feasible at any κ there is a smallest such κ_0 , and feasibility holds on $[\kappa_0, \bar{\kappa}]$, where $\bar{\kappa}$ is the value at which 0 leaves the interval. At κ_0 the posterior means coincide with those induced by the threshold signal of the previous case, so the bias carries over by continuity. For $\kappa \in (\kappa_0, \bar{\kappa})$ the formula $P_a = \frac{1}{2} + \frac{\kappa}{2} \frac{x_a + x_b}{x_b - x_a}$ applies, and since $x_a + x_b = 2\frac{x_a + x_b}{2} > 0$ the vote share increases in κ .

For $\kappa \geq \bar{\kappa}$ we have $0 \notin [\theta_a, \theta_b]$, voters acquire no information, and all vote for the ex-ante more attractive party, which is party a . So $P_a = 1$, the maximal bias. $\square$

A.3.4 Lemma 4

Proof. Let ρ be supported on the line through the origin with direction $A(x_b - x_a)$. We first show that the objectives of (Pn) and (P1) agree up to a positive factor. Write $e = \frac{A(x_b - x_a)}{\|A(x_b - x_a)\|}$, so that $\theta \sim \rho$ can be represented as $\theta = \hat{\theta} e$ with $\hat{\theta} \sim \hat{\rho}$. Then $\theta^\top \theta = \hat{\theta}^2$ and

$$(x_b - x_a)^\top A(\theta - \frac{x_a + x_b}{2}) = \|A(x_b - x_a)\| \left(\hat{\theta} - \frac{\hat{x}_a + \hat{x}_b}{2} \right) = \frac{\|A(x_b - x_a)\|^2}{(x_b - x_a)^\top A(x_b - x_a)} (\hat{x}_b - \hat{x}_a) \left(\hat{\theta} - \frac{\hat{x}_a + \hat{x}_b}{2} \right),$$

where the last step uses $\hat{x}_b - \hat{x}_a = \frac{(x_b - x_a)^\top A(x_b - x_a)}{\|A(x_b - x_a)\|}$. Dividing the objective of (Pn) by the positive constant $\frac{\|A(x_b - x_a)\|^2}{(x_b - x_a)^\top A(x_b - x_a)}$ therefore yields the objective of (P1), with cost coefficient $\hat{\kappa}$ as in (14). Since the constant is positive, the two problems have the same maximizers.

For the constraint, we use that for ρ supported on the line through the origin with direction $A(x_b - x_a)$,

$$\rho \preceq_{\text{MPS}} \mu \iff \hat{\rho} \preceq_{\text{MPS}} \hat{\mu}.$$

The only if direction holds because the mean-preserving contraction order is preserved under scalar projection. For the if direction, let $\text{Proj}_{0, A(x_b - x_a)} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be the orthogonal projection onto the line through the origin with direction $A(x_b - x_a)$, extended to distributions via the pushforward. Note this is different from the scalar projection $\text{proj}_{A(x_b - x_a)}$. Since ρ is supported on that line, $\hat{\rho} \preceq_{\text{MPS}} \hat{\mu}$ means $\rho \preceq_{\text{MPS}} \text{Proj}_{0, A(x_b - x_a)}(\mu)$. Because μ is spherical, $\text{Proj}_{0, A(x_b - x_a)}(\mu) \preceq_{\text{MPS}} \mu$, and transitivity gives $\rho \preceq_{\text{MPS}} \mu$.

Combining the two equivalences, among distributions ρ supported on the line through the origin with direction $A(x_b - x_a)$, ρ solves (Pn) if and only if $\hat{\rho}$ solves (P1). This establishes the if direction. For the only if direction, Theorem 1 implies that any solution to (Pn) is supported on that line, and the equivalence just shown then implies that $\hat{\rho}$ solves (P1). $\square$

A.3.5 Proposition 1

Proof. By Lemma 4, the optimal ρ projects to the optimal $\hat{\rho}$ of (P1), a one-dimensional voter learning problem with cost $\hat{\kappa}$, party positions $\hat{x}_a, \hat{x}_b$, and prior $\hat{\mu}$. Moreover the vote shares from (Pn) equal the one-dimensional vote shares for (P1), and the share of voters whose true ideal point is A -closer to x_a equals $F_{\hat{\mu}}(\frac{\hat{x}_a + \hat{x}_b}{2})$.

The assumption $x_a^\top A x_a < x_b^\top A x_b$ gives $|\hat{x}_a| < |\hat{x}_b|$: with $d = x_b - x_a$,

$$|\hat{x}_a| < |\hat{x}_b| \iff (d^\top A x_b)^2 - (d^\top A x_a)^2 > 0 \iff (d^\top A(x_b + x_a))(d^\top A(x_b - x_a)) > 0.$$

The first factor equals $x_b^\top A x_b - x_a^\top A x_a > 0$ by assumption, and the second equals $d^\top A d > 0$ since $x_a \neq x_b$. Corollary 4 applied to (P1) then gives $P_a > F_{\hat{\mu}}(\frac{\hat{x}_a + \hat{x}_b}{2})$, which is the claimed bias, and shows that P_a increases in $\hat{\kappa}$ and decreases in $|\hat{x}_b - \hat{x}_a|$, holding $\frac{\hat{x}_a + \hat{x}_b}{2}$ fixed.

It remains to translate the comparative statics in Corollary 4 to the Proposition 1. Increasing κ is equivalent to increasing $\hat{\kappa}$ by (14), which weakly increases $P_a(x_a, x_b)$. Scaling up $x_b - x_a$ while holding $\frac{x_a + x_b}{2}$ fixed is equivalent to scaling up $\hat{x}_b - \hat{x}_a$ while holding $\frac{\hat{x}_a + \hat{x}_b}{2}$ and $\hat{\kappa}$ fixed, which weakly decreases $P_a(x_a, x_b)$. $\square$

B Appendix: Electoral Competition

B.1 One-Dimensional Policy Space

We first solve the electoral-competition game in the one-dimensional case $n = 1$, in which party platforms are scalars $x_a, x_b \in \mathbb{R}$ and the voter prior $\mu \in \Delta(\mathbb{R})$ has a log-concave density f_μ that is symmetric around 0. Recall that the definition (6) of the posterior means from the multidimensional case simplifies in one dimension to

$$\begin{aligned}\theta_a(x_a, x_b) &:= \frac{x_a + x_b}{2} - \frac{x_b - x_a}{2\kappa}, \\ \theta_b(x_a, x_b) &:= \frac{x_a + x_b}{2} + \frac{x_b - x_a}{2\kappa}.\end{aligned}\tag{24}$$

The following terminology is used in the following.

Definition 2. *We say*

- **feasibility holds at (x_a, x_b)** if $\{\theta_a(x_a, x_b), \theta_b(x_a, x_b)\}$ is feasible.
- **feasibility is slack at (x_a, x_b)** if feasibility holds on an open neighborhood of (x_a, x_b) .

We use slack feasibility in the best-response lemma below. Lemma 3 gives a useful sufficient condition for feasibility holding at (x_a, x_b) . The following strengthening of Lemma 3 gives a useful sufficient condition for feasibility being slack at (x_a, x_b) .

Lemma 5. *If $|x_b - x_a| < \kappa\Delta$ and $\theta_a(x_a, x_b) < 0 < \theta_b(x_a, x_b)$, then feasibility is slack at (x_a, x_b) .*

Proof. Let $|x_b - x_a| < \kappa\Delta$ and $\theta_a(x_a, x_b) < 0 < \theta_b(x_a, x_b)$. By Lemma 3 and $\theta_a(x_a, x_b)$ and $\theta_b(x_a, x_b)$ being continuous in x_a and x_b , then feasibility holds on an open neighborhood of (x_a, x_b) . $\square$

Lemma 6. *Let $\kappa < 1$ and $x_a \leq x'_a \leq x'_b \leq x_b$, and suppose feasibility holds at (x_a, x_b) . If $\theta_a(x'_a, x'_b) \leq 0 \leq \theta_b(x'_a, x'_b)$, then feasibility holds at (x'_a, x'_b) .*

Proof. For $\kappa < 1$, the posterior mean $\theta_a(x_a, x_b) = \frac{x_a + x_b}{2} - \frac{x_b - x_a}{2\kappa}$ increases in x_a and decreases in x_b , and $\theta_b(x_a, x_b) = \frac{x_a + x_b}{2} + \frac{x_b - x_a}{2\kappa}$ decreases in x_a and increases in x_b . Therefore

$$\theta_a(x_a, x_b) \leq \theta_a(x'_a, x'_b) \leq \theta_b(x'_a, x'_b) \leq \theta_b(x_a, x_b).$$

By feasibility at (x_a, x_b) , the mean-zero distribution on $\{\theta_a(x_a, x_b), \theta_b(x_a, x_b)\}$ is a mean-preserving contraction of μ . By $\theta_a(x'_a, x'_b) \leq 0 \leq \theta_b(x'_a, x'_b)$ and the inclusion above, the mean-zero distribution on $\{\theta_a(x'_a, x'_b), \theta_b(x'_a, x'_b)\}$ is a mean-preserving contraction of the mean-zero distribution on $\{\theta_a(x_a, x_b), \theta_b(x_a, x_b)\}$, hence of μ . $\square$

B.1.1 Lemmas on Party Positions

The following lemmas describes properties of any equilibrium. Without loss, assume $x_a^* \leq x_b^*$.

Lemma 7 (Platform Proximity). *In any equilibrium, each party's platform is weakly closer to its own ideal policy than the opponent's platform, that is, $|x_j - x_j^*| \leq |x_{-j} - x_j^*|$ for $j \in \{a, b\}$.*

Proof. Recall that we restrict attention to equilibria where both parties obtain positive vote shares. Suppose $|x_j - x_j^*| > |x_{-j} - x_j^*|$ for some $j \in \{a, b\}$, so $u(x_j, x_j^*) < u(x_{-j}, x_j^*)$. Because party j wins with positive probability, its expected utility is strictly less than $u(x_{-j}, x_j^*)$. By deviating to the platform x_{-j} , policy x_{-j} is implemented with certainty and obtains party j obtains utility $u(x_{-j}, x_j^*)$. The deviation is profitable, so (x_a, x_b) is not an equilibrium. $\square$

Lemma 8 (Equilibrium Bounds). *If $x_a^* < x_b^*$, then in any equilibrium:*

- (1) $x_a \leq x_b$.
- (2) If $x_a^* \leq 0 \leq x_b^*$, then $2x_a^* \leq x_a \leq x_b^*$ and $x_a^* \leq x_b \leq 2x_b^*$.
- (3) If $0 < x_a^*$, then either $x_a = x_b$ or $x_a^* \leq x_a < x_b \leq 2x_b^* - x_a^*$. An analogous statement holds if $x_b^* < 0$.

In every case, $|x_b - x_a| \leq 2|x_b^ - x_a^*|$.*

Proof. (1) Suppose, for the sake of contradiction, that $x_a > x_b$. The inequality $|x_a - x_a^*| \leq |x_b - x_a^*|$ from Lemma 7 together with $x_a > x_b$ implies $\frac{x_a + x_b}{2} \leq p$. By the same argument with the roles of party a and b exchanged, $x_b^* \leq \frac{x_a + x_b}{2}$. Together, $x_b^* \leq x_a^*$, contradicting $x_a^* < x_b^*$.

(2) First we show $x_a \leq x_b^*$. Suppose $x_a > x_b^*$. Lemma 7 for party b gives $x_b \leq x_a$, so $x_a = x_b$ by (1). For small $\varepsilon > 0$, both posterior means $\theta_a(x_a, x_b), \theta_b(x_a, x_b)$ at $(x_a - \varepsilon, x_a)$ are positive, so deviating to $x_a - \varepsilon$ gives party a all votes by Corollary 3 and a better policy by $x_a^* \leq 0$, a contradiction. By symmetry of the setup, an analogous argument gives $x_b \geq x_a^*$.

Next we show $2x_a^* \leq x_a$. Suppose $x_a < 2x_a^*$, so $x_a < x_a^* \leq 0$. We distinguish three cases.

If $x_b \leq 0$, then Lemma 7 and $x_a^* \leq x_b$ give $x_a \geq 2x_a^* - x_b \geq 2x_a^*$, a contradiction.

If $x_b > 0$ and $|x_a| \geq |x_b|$, then $P_a(x_a, x_b) \leq \frac{1}{2} \leq P_a(0, x_b)$ and $u(x_b, x_a^*) \leq u(x_a, x_a^*) < u(0, x_a^*)$ by Corollary 3, Lemma 7, and $x_a < 2x_a^*$. Together, $U_a(x_a, x_b) \leq \frac{1}{2}u(x_a, x_a^*) + \frac{1}{2}u(x_b, x_a^*) < U_a(0, x_b)$, so 0 is a profitable deviation.

If $x_b > 0$ and $|x_a| < |x_b|$, fix $\varepsilon > 0$ with $x_a + \varepsilon < x_a^*$, so $|x_a + \varepsilon| < |x_b|$. By $u(x_a + \varepsilon, x_a^*) > u(x_a, x_a^*) \geq u(x_b, x_a^*)$, where the second inequality holds by Lemma 7, the deviation to $x_a + \varepsilon$ is profitable if $P_a(x_a + \varepsilon, x_b) \geq P_a(x_a, x_b)$, which we show now. If feasibility holds at (x_a, x_b) , then $P_a(x_a, x_b) = \frac{1}{2} + \frac{\kappa}{2} \frac{x_b + x_a}{x_b - x_a} < 1$, which is increasing in x_a by $x_b > 0$, and Corollary 3 gives $P_a(x_a + \varepsilon, x_b) = 1$ or $P_a(x_a + \varepsilon, x_b) \geq \frac{1}{2} + \frac{\kappa}{2} \frac{x_b + (x_a + \varepsilon)}{x_b - (x_a + \varepsilon)}$. If feasibility does not hold at (x_a, x_b) , then $0 \in (\theta_a, \theta_b)$, since party a would otherwise obtain all votes. Both properties persist at $(x_a + \varepsilon, x_b)$ for small ε , because the posterior means are continuous in x_a and the set of feasible pairs is closed. By Proposition 2 and Corollary 3, a threshold signal is optimal at both position pairs, with thresholds $t, t_\varepsilon > \frac{x_a + x_b}{2} > x_a$ and vote shares $F_\mu(t), F_\mu(t_\varepsilon)$. The voter objective $b(t) - \kappa c(t)$ is supermodular in (x_a, t) on the region $t > x_a$, so $t_\varepsilon \geq t$ and $P_a(x_a + \varepsilon, x_b) \geq P_a(x_a, x_b)$. Both the vote share and the policy utility when winning increase when raising x_a , so $x_a < x_a^*$ is suboptimal.

Analogously, $x_b \leq 2x_b^*$.

(3) Let $0 < x_a^*$ and $x_a \neq x_b$, so $x_a < x_b$ by (1).

First we show $x_a^* \leq x_a$. Suppose $x_a < x_a^*$. If $x_b < x_a^*$, then $x_a \geq x_b$ by Lemma 7 contradicting $x_a < x_b$. If $x_b \geq x_a^*$ and $|x_a| \geq |x_b|$, then $x_a \leq -x_b$, so x_a is further from x_a^* than x_b contradicting Lemma 7. If $x_b \geq x_a^*$, $|x_a| \leq |x_b|$ and $x_a < x_a^*$, the third case in the proof of (2) applies verbatim and yields a profitable deviation via increasing x_a , a contradiction.

Next, we show $x_a \leq x_b^*$. If not, Lemma 7 for party b gives $x_b \leq x_a$, contradicting $x_a < x_b$.

Finally, by Lemma 7 for party b and $x_a^* \leq x_a \leq x_b^*$, we have $x_b - x_b^* \leq |x_a - x_b^*| = x_b^* - x_a \leq x_b^* - x_a^*$, so $x_b \leq 2x_b^* - x_a^*$.

The statement for $x_b^* < 0$ follows analogously. $\square$

Lemma 9. *If $x_a^* = x_b^*$, then $x_a = x_b$ in any equilibrium.*

Proof. Suppose $x_a \neq x_b$ and write $x^* := x_a^* = x_b^*$. Lemma 7 applied to both parties gives $|x_a - x^*| = |x_b - x^*| =: d$, so $d > 0$ and $x^* = \frac{x_a + x_b}{2}$. The implemented policy is at distance d from x^* regardless of the election outcome, so each party's expected utility equals $-d^2$. Let j be a party with $|x_j| = \min\{|x_a|, |x_b|\}$ and consider a deviation of party j to x^* . By $x_a \neq x_b$, $|x^*| < |x_{-j}|$, and the midpoint of x^* and x_{-j} lies strictly on the same side of 0 as x_{-j} . By Corollary 4, party j 's vote share P_j at the deviation weakly exceeds the share of voters whose true ideal point is closer to x^* than to x_{-j} , and that share is at least $\frac{1}{2}$ by the previous sentence. Party j 's expected utility from the deviation is at least $1/2 \cdot 0 + 1/2 \cdot (-d^2) > -d^2$, so (x_a, x_b) is not an equilibrium. $\square$

Lemma 10. *If $x_a^* \leq 0 \leq x_b^*$ and $\kappa \geq 1$, then the unique equilibrium is $(x_a, x_b) = (0, 0)$.*

Proof. Under $\kappa \geq 1$, the implemented policy is zero if one party chooses policy 0. This follows from Corollary 3 and the fact that $\theta_a(x_a, x_b)$ and $\theta_b(x_a, x_b)$ are weakly between x_a and x_b for $\kappa \geq 1$. Therefore, $(x_a, x_b) = (0, 0)$ is an equilibrium.

By Lemma 8 (1) and Lemma 9, $x_a \leq x_b$ in any equilibrium.

Suppose (x_a, x_b) is an equilibrium. The average implemented policy is weakly positive or weakly negative. Suppose without loss that it is weakly positive. Then party a can obtain policy 0 for certain by choosing $x_a = 0$. This weakly improves the utility of party a because the policy outcome is weakly better in expectation and has no variance. Thus, if the policy outcome under (x_a, x_b) is not 0 for certain, then this is a profitable deviation. Finally, if the policy outcome is 0 for certain, then at least one party must choose position 0. The other party must also choose position 0, otherwise only one party would obtain a positive vote share. $\square$

B.1.2 Best Response Lemma

Next, we provide a lemma that partially characterizes parties best responses. We need a few definitions. Define

$$\tilde{P}_a(x_a, x_b) := \frac{1}{2} + \frac{\kappa}{2} \frac{x_b + x_a}{x_b - x_a}, \quad (25)$$

$$\tilde{U}_a(x_a, x_b) := \tilde{P}_a(x_a, x_b)u(x_a, x_a^*) + (1 - \tilde{P}_a(x_a, x_b))u(x_b, x_a^*). \quad (26)$$

We call $\tilde{P}_a$ and $\tilde{U}_a$ the **pseudo vote share** and **pseudo utility**, respectively, because they are equal to the vote share and expected utility of party a only if feasibility holds at (x_a, x_b) , by Corollary 3.

Lemma 11 (Best Response). *Let $x_a^* < x_b^*$, $\kappa < 1$, and $x_b \in \mathbb{R}$. Define*

$$\hat{x}_a = x_a^* + \frac{\kappa}{1 - \kappa} x_b.$$

Necessity: *If x_a is a best response of party a to x_b and feasibility is slack at (x_a, x_b) , then $x_a = \hat{x}_a$.*

Sufficiency: *If feasibility holds at $(\hat{x}_a, x_b)$ and at $(-x_b, x_b)$, then $\hat{x}_a$ is a best response of party a to x_b .*

Analogous statements hold when exchanging a and b (maintaining $x_a^ < x_b^*$).*

Proof. Necessity: If feasibility is slack at (x_a, x_b) , then $U_a(x_a, x_b)$ is locally given by the pseudo utility

$$\begin{aligned} \tilde{U}_a(x_a, x_b) &= \tilde{P}_a(x_a, x_b)u(x_a, x_a^*) + (1 - \tilde{P}_a(x_a, x_b))u(x_b, x_a^*) \\ &= (u(x_a, x_a^*) - u(x_b, x_a^*))\tilde{P}_a(x_a, x_b) + u(x_b, x_a^*) \\ &= 2 \left(x_a^* - \frac{x_a + x_b}{2} \right) (x_a - x_b) \left(\frac{1}{2} + \frac{\kappa}{2} \frac{x_a + x_b}{x_b - x_a} \right) + u(x_b, x_a^*) \\ &= \left(\frac{x_a + x_b}{2} - x_a^* \right) (x_b(1 + \kappa) - x_a(1 - \kappa)) + u(x_b, x_a^*). \end{aligned} \quad (27)$$

Since the domain of x_a is unbounded, if x_a is a best response, then it satisfies the first-order condition $\frac{d\tilde{U}_a(x_a, x_b)}{dx_a} = 0$. The unique solution to the first-order condition is

$$\hat{x}_a = x_a^* + \frac{\kappa}{1 - \kappa} x_b.$$

Sufficiency: Inserting $\hat{x}_a$ into (27) gives

$$U_a(\hat{x}_a, x_b) = \frac{1 - \kappa}{2} \left(\frac{x_b}{1 - \kappa} - x_a^* \right)^2 + u(x_b, x_a^*) \geq u(x_b, x_a^*), \quad (28)$$

with equality if and only if $x_b = (1 - \kappa)x_a^*$. Take any $x_a \neq \hat{x}_a$.

If x_a is at least as far from x_a^* as x_b is, then $U_a(x_a, x_b) \leq u(x_b, x_a^*)$, which is less or equal to $U_a(\hat{x}_a, x_b)$ by (28). Assume for the rest of the proof that x_a is strictly closer to x_a^* than x_b is.

If feasibility holds at (x_a, x_b) , then $U_a(x_a, x_b) = \tilde{U}_a(x_a, x_b) < \tilde{U}_a(\hat{x}_a, x_b) = U_a(\hat{x}_a, x_b)$ by strict concavity, since $x_a \neq \hat{x}_a$ and $\hat{x}_a$ is the unique maximizer of the pseudo utility.

If feasibility does not hold at (x_a, x_b) , then by Proposition 2, either the origin does not lie between $\theta_a(x_a, x_b)$ and $\theta_b(x_a, x_b)$, or the voter uses a threshold signal. If the origin does not lie between $\theta_a(x_a, x_b)$ and $\theta_b(x_a, x_b)$, party a obtains either no votes or all votes. With no votes, $U_a(x_a, x_b) = u(x_b, x_a^*) \leq U_a(\hat{x}_a, x_b)$. With all votes, both posterior means lie on the same side of the origin with $\theta_a(x_a, x_b)$ closer to 0 than $\theta_b(x_a, x_b)$, so $\tilde{P}_a(x_a, x_b) > 1$. Together with x_a being closer to x_a^* than x_b ,

$$U_a(x_a, x_b) = u(x_a, x_a^*) \leq \tilde{U}_a(x_a, x_b) < \tilde{U}_a(\hat{x}_a, x_b) = U_a(\hat{x}_a, x_b).$$

If the voter uses a threshold signal, the origin lies between $\theta_a(x_a, x_b)$ and $\theta_b(x_a, x_b)$, and we claim $|x_a| > |x_b|$. Suppose not, so $|x_a| \leq |x_b|$. Writing $x_b > 0$ and $x_a < x_b$ without loss gives $-x_b \leq x_a < x_b$. By feasibility at $(-x_b, x_b)$ and Lemma 6, feasibility holds at (x_a, x_b) , a contradiction. Thus $|x_a| > |x_b|$, and by Corollary 3(2), $P_a(x_a, x_b) < \tilde{P}_a(x_a, x_b)$. Since x_a is closer to x_a^* than x_b is,

$$U_a(x_a, x_b) \leq \tilde{U}_a(x_a, x_b) < \tilde{U}_a(\hat{x}_a, x_b) = U_a(\hat{x}_a, x_b).$$

In every case $U_a(x_a, x_b) \leq U_a(\hat{x}_a, x_b)$. Hence $\hat{x}_a$ is a best response.

By symmetry, in particular of the prior around 0, analogous statements hold when exchanging a and b . $\square$

B.1.3 Equilibrium

We first describe equilibrium in a symmetric setup and then in an asymmetric setup. Define $\underline{\kappa}$ by

$$\frac{1 - \underline{\kappa}}{\underline{\kappa}} |x_a^*| = \mathbb{E}[|\theta|].$$

The parameter $\underline{\kappa}$ is the smallest information cost parameter for which feasibility holds at the equilibrium positions (x_a, x_b) according to the following proposition.

Proposition 3. *Let $x_a^* < 0 < x_b^* = -x_a^*$. The following constitutes an equilibrium.*

	Party platforms	Revealed voter ideology
$\kappa \geq 1$:	Downsian equilibrium $(x_a, x_b) = (0, 0)$	$\rho = \delta(0)$
$\kappa \in (\underline{\kappa}, 1)$:	Polarizing equilibrium $(x_a, x_b) = (1 - \kappa)(x_a^*, x_b^*)$	$\rho = \frac{1}{2}\delta\left(\frac{x_a}{\kappa}\right) + \frac{1}{2}\delta\left(\frac{x_b}{\kappa}\right)$

The equilibrium platforms are unique if $|x_b^ - x_a^*| < \kappa \mathbb{E}[|\theta|]$ and $\kappa \neq 1/2$.*

Proof. Lemma 10 shows that the unique equilibrium is $(x_a, x_b) = (0, 0)$ if $\kappa \geq 1$. Let $\kappa < 1$.

Equilibrium verification: The positions $(x_a, x_b) = (1 - \kappa)(x_a^*, x_b^*)$ solve $x_a = x_a^* + \frac{\kappa}{1 - \kappa} x_b$ and $x_b = x_b^* + \frac{\kappa}{1 - \kappa} x_a$. By $\kappa > \underline{\kappa}$, feasibility holds at (x_a, x_b) . By $\kappa < 1$, $|x_a| = |x_b|$, and the sufficiency part of Lemma 11, this constitutes mutual best responses and hence an equilibrium.

Uniqueness: By Lemma 8 and $|x_b^* - x_a^*| < \kappa \mathbb{E}[|\theta|]$, in any equilibrium (x_a, x_b) ,

$$|x_b - x_a| \leq 2|x_b^* - x_a^*| < 2\kappa \mathbb{E}[|\theta|] = \kappa \Delta.$$

By Lemma 5, either $0 \notin (\theta_a(x_a, x_b), \theta_b(x_a, x_b))$ or feasibility is slack at (x_a, x_b) . In the former case, one party obtains all votes or $x_a = x_b$. Let $x_a = x_b \geq 0$, the other case being analogous. By Corollary 3, party a obtains positive votes by deviating to $x_a - \varepsilon$ for ε small enough, so $x_a - \varepsilon$ is a profitable deviation. Finally, let feasibility be slack at (x_a, x_b) . By $\kappa \in (0, 1)$ and the necessity part of Lemma 11, then $x_a = x_a^* + \frac{\kappa}{1-\kappa}x_b$ and $x_b = x_b^* + \frac{\kappa}{1-\kappa}x_a$. Solving, we obtain

$$(x_a, x_b) = (1 - \kappa)(x_a^*, x_b^*) \quad (29)$$

if $\kappa \neq 1/2$.¹⁰ □

Proposition 4. *Suppose $x_a^* < 0 < x_b^*$ and $|x_a^*| < |x_b^*|$. The positions*

$$\begin{aligned} x_a(\kappa) &= \frac{1 - \kappa}{1 - 2\kappa} \left((1 - \kappa)x_a^* + \kappa x_b^* \right) \\ x_b(\kappa) &= \frac{1 - \kappa}{1 - 2\kappa} \left((1 - \kappa)x_b^* + \kappa x_a^* \right) \end{aligned}$$

constitute an equilibrium for

$$\kappa \in \left(0, \frac{x_b^* - x_a^*}{3x_b^* - x_a^*} \right) \cup \left(\frac{x_b^* - x_a^*}{x_b^* - 3x_a^*}, 1 \right) \quad \text{and} \quad \max\{|x_a(\kappa)|, |x_b(\kappa)|\} < \kappa \mathbb{E}[|\theta|]. \quad (30)$$

These are the unique equilibrium platforms for κ satisfying (30) if additionally $|x_b^ - x_a^*| < \kappa \mathbb{E}[|\theta|]$.*

If $\kappa \geq 1$, the unique equilibrium platforms are $(x_a, x_b) = (0, 0)$.

Proof. If $\kappa \geq 1$, the unique equilibrium is $(x_a, x_b) = (0, 0)$ by Lemma 10. Assume $\kappa < 1$ henceforth. Imposing the best-response conditions from Lemma 11,

$$\begin{aligned} x_a &= x_a^* + \frac{\kappa}{1 - \kappa}x_b, \\ x_b &= x_b^* + \frac{\kappa}{1 - \kappa}x_a, \end{aligned} \quad (31)$$

and solving yields the unique solution $(\tilde{x}_a, \tilde{x}_b) = (x_a(\kappa), x_b(\kappa))$ for $\kappa \neq 1/2$. We first verify that $(\tilde{x}_a, \tilde{x}_b)$ constitutes an equilibrium under (30), then establish uniqueness.

Equilibrium verification: By the sufficiency part of Lemma 11, $(\tilde{x}_a, \tilde{x}_b)$ constitutes mutual best responses whenever feasibility holds at $(\tilde{x}_a, \tilde{x}_b)$, $(-\tilde{x}_b, \tilde{x}_b)$, and $(\tilde{x}_a, -\tilde{x}_a)$. Take $|\tilde{x}_b| \geq |\tilde{x}_a|$, the other case being analogous.

Since $\max\{|\tilde{x}_a|, |\tilde{x}_b|\} < \kappa \mathbb{E}[|\theta|]$, we have $|\tilde{x}_b - (-\tilde{x}_b)| = 2|\tilde{x}_b| < 2\kappa \mathbb{E}[|\theta|] = \kappa \Delta$. Together with $\theta_a(-|\tilde{x}_b|, |\tilde{x}_b|) \leq 0 \leq \theta_b(-|\tilde{x}_b|, |\tilde{x}_b|)$ and Lemma 3, feasibility holds at $(-\tilde{x}_b, \tilde{x}_b)$. By $|\tilde{x}_a| \leq |\tilde{x}_b|$, the platforms of both $(\tilde{x}_a, \tilde{x}_b)$ and $(\tilde{x}_a, -\tilde{x}_a)$ lie in $[-|\tilde{x}_b|, |\tilde{x}_b|]$, so by Lemma 6 feasibility holds at

¹⁰If $\kappa = 1/2$, the best response function has slope 1 and all (x_a, x_b) with $x_b = x_a + \frac{1}{2}(x_b^* - x_a^*)$ solve the equations $x_a = x_a^* + \frac{\kappa}{1-\kappa}x_b$ and $x_b = x_b^* + \frac{\kappa}{1-\kappa}x_a$.

either pair once the origin lies between its posterior means θ_a, θ_b . The posterior means at $(\tilde{x}_a, -\tilde{x}_a)$ equal $\pm|\tilde{x}_a|/\kappa$, so the origin lies between them and feasibility holds at $(\tilde{x}_a, -\tilde{x}_a)$. For $(\tilde{x}_a, \tilde{x}_b)$ the same conclusion follows once $\theta_a(\tilde{x}_a, \tilde{x}_b) \leq 0 \leq \theta_b(\tilde{x}_a, \tilde{x}_b)$, so establishing that inequality suffices to conclude that feasibility holds at $(\tilde{x}_a, \tilde{x}_b)$.

Under $\kappa > 1/2$, $\tilde{x}_a$ is the more extreme platform by $|x_a^*| < |x_b^*|$, so $\theta_a(\tilde{x}_a, \tilde{x}_b) < 0$ automatically, and $\theta_a(\tilde{x}_a, \tilde{x}_b) \leq 0 \leq \theta_b(\tilde{x}_a, \tilde{x}_b)$ holds if

$$\theta_b(\tilde{x}_a, \tilde{x}_b) = \frac{\tilde{x}_a + \tilde{x}_b}{2} + \frac{\tilde{x}_b - \tilde{x}_a}{2\kappa} > 0 \iff \kappa > \frac{x_b^* - x_a^*}{x_b^* - 3x_a^*} > \frac{1}{2}.$$

Under $\kappa < 1/2$, $\tilde{x}_b$ is the more extreme platform by $|x_a^*| < |x_b^*|$, so $\theta_b(\tilde{x}_a, \tilde{x}_b) > 0$ automatically, and $\theta_a(\tilde{x}_a, \tilde{x}_b) \leq 0 \leq \theta_b(\tilde{x}_a, \tilde{x}_b)$ holds if

$$\theta_a(\tilde{x}_a, \tilde{x}_b) = \frac{\tilde{x}_a + \tilde{x}_b}{2} - \frac{\tilde{x}_b - \tilde{x}_a}{2\kappa} < 0 \iff \kappa < \frac{x_b^* - x_a^*}{3x_b^* - x_a^*} < \frac{1}{2}.$$

The bounds hold by (30).

Uniqueness: The argument follows the proof of Proposition 3, using that $(\tilde{x}_a, \tilde{x}_b)$ is the unique solution to (31). $\square$

B.2 Multidimensional Policy Space

We prove Theorems 2 and 3 in parallel through several steps, building on our results for a one-dimensional policy space. In Section B.2.1 we show how the party objective relates to the objective under a one-dimensional policy space. In Section B.2.2 we combine this reduction with the one-dimensional best response Lemma 11 to prove a best response lemma for the multidimensional policy space. In Section B.2.3 we verify that our equilibrium candidate is an equilibrium, using the sufficiency part of the best response lemma. In Section B.2.4 we prove uniqueness. Finally, in Section B.2.5 we treat the case $\kappa \geq 1$.

We set $A = I_n$, the assumption maintained in the electoral competition section. Then, in the reduction of the voter learning problem to one dimension (P1), $\hat{\kappa} = \kappa$, and $\text{proj}_{A(x_b - x_a)} = \text{proj}_{x_b - x_a}$ is the scalar projection onto the line through the origin with direction $x_b - x_a$. We generalize Definition 1 and 2 verbatim to the multidimensional policy space using the definition of $\theta_a(x_a, x_b)$ and $\theta_b(x_a, x_b)$ from (6).

B.2.1 Reduction to One Dimension

We now show that confining both parties to a common line reduces the party game to a one-dimensional game. Define $\hat{x}_a, \hat{x}_b$ as in (13) with $A = I_n$ and

$$\hat{x}_a^* = \text{proj}_{x_b - x_a}(x_a^*), \quad \hat{x}_b^* = \text{proj}_{x_b - x_a}(x_b^*).$$

Define $\hat{P}_a(x_a, x_b)$ as the vote share for party a under optimal learning in the one-dimensional model with party positions $\hat{x}_a, \hat{x}_b$ and voter prior $\hat{\mu} = \text{proj}_{x_b - x_a}(\mu)$. Party a 's objective under a

one-dimensional policy space is

$$\widehat{U}_a(\hat{x}_a, \hat{x}_b) = \widehat{P}_a(\hat{x}_a, \hat{x}_b) \widehat{U}(\hat{x}_a, \hat{x}_a^*) + (1 - \widehat{P}_a(\hat{x}_a, \hat{x}_b)) \widehat{U}(\hat{x}_b, \hat{x}_a^*),$$

where $\widehat{U}(x, y) = -(x - y)^2$. Define $\widehat{U}_b$ analogously. Let D_a^2 be the squared Euclidean distance of x_a^* from the line through x_a and x_b . The next lemma shows that the party objective coincides with the one-dimensional objective up to the additive constant D_a^2 , which depends on x_a, x_b only through the line they span. The result follows from the one-dimensional reduction of voter learning and parties' quadratic preferences.

Lemma 12. *Party a's utility satisfies*

$$U_a(x_a, x_b) = \widehat{U}_a(\hat{x}_a, \hat{x}_b) - D_a^2 = \widehat{P}_a(\hat{x}_a, \hat{x}_b) 2(\hat{x}_a - \hat{x}_b) \left(\hat{x}_a^* - \frac{\hat{x}_a + \hat{x}_b}{2} \right) + u(x_b, x_a^*).$$

The analogous identity holds after exchanging a and b.

Proof. By Lemma 4, $P_a(x_a, x_b) = \widehat{P}_a(\hat{x}_a, \hat{x}_b)$. Hence

$$\begin{aligned} U_a(x_a, x_b) &= P_a(x_a, x_b) (u(x_a, x_a^*) - u(x_b, x_a^*)) + u(x_b, x_a^*) \\ &= \widehat{P}_a(\hat{x}_a, \hat{x}_b) 2(x_a - x_b)^\top \left(x_a^* - \frac{x_a + x_b}{2} \right) + u(x_b, x_a^*) \\ &= \widehat{P}_a(\hat{x}_a, \hat{x}_b) 2(\hat{x}_a - \hat{x}_b) \left(\hat{x}_a^* - \frac{\hat{x}_a + \hat{x}_b}{2} \right) + \widehat{U}(\hat{x}_b, \hat{x}_a^*) - D_a^2 \\ &= \widehat{U}_a(\hat{x}_a, \hat{x}_b) - D_a^2, \end{aligned}$$

where the third line uses Pythagoras. □

In particular, any equilibrium (x_a, x_b) remains an equilibrium when both parties are restricted to the line through x_a and x_b , where it is an equilibrium of the corresponding one-dimensional game.

B.2.2 Multidimensional Best Response Lemma

The following lemma on best responses exploits the reduction to one dimension proven by the above lemma. Following definition (5), $\text{Proj}_{(1-\kappa)x_a^*, x_b}$ denotes the orthogonal projection of $\mathbb{R}^n$ onto the line through $(1 - \kappa)x_a^*$ and x_b .

As under one dimension, we define the pseudo vote-share and pseudo utility as the vote share and utility when feasibility holds at (x_a, x_b) :

$$\begin{aligned} \widetilde{P}_a(x_a, x_b) &:= \frac{1}{2} + \frac{\kappa}{2} \frac{\hat{x}_a + \hat{x}_b}{\hat{x}_b - \hat{x}_a}, \\ \widetilde{U}_a(x_a, x_b) &:= \widetilde{P}_a(x_a, x_b) u(x_a, x_a^*) + (1 - \widetilde{P}_a(x_a, x_b)) u(x_b, x_a^*). \end{aligned}$$

Lemma 13 (Multidimensional Best Response Lemma). *Let $\kappa < 1$, $x_b \in \mathbb{R}^n$. Define*

$$\tilde{x}_a := \text{Proj}_{(1-\kappa)x_a^*, x_b} \left(x_a^* + \frac{\kappa}{1 - \kappa} x_b \right).$$

Necessity: If x_a is the best response of party a to x_b and feasibility is slack at (x_a, x_b) , then $x_a = \tilde{x}_a$.

Sufficiency: If feasibility holds at $(\tilde{x}_a, x_b)$ and at $(-x_b, x_b)$, then $\tilde{x}_a$ is the best response to x_b .

Proof. Necessity: Suppose x_a is a best response to x_b and feasibility is slack at (x_a, x_b) . By Lemma 12,

$$U_a(x_a, x_b) = \hat{P}_a(\hat{x}_a, \hat{x}_b)2(\hat{x}_a - \hat{x}_b)\left(\hat{x}_a^* - \frac{\hat{x}_a + \hat{x}_b}{2}\right) + u(x_b, x_a^*),$$

where $\hat{x}_a, \hat{x}_b, \hat{x}_a^*$ are the scalar projections of x_a, x_b, x_a^* on the line through x_a and x_b . If feasibility is slack at (x_a, x_b) , then $U_a(x_a, x_b)$ is locally given by the pseudo utility

$$\begin{aligned} \tilde{U}_a(x_a, x_b) &= \tilde{P}_a(x_a, x_b)2(\hat{x}_a - \hat{x}_b)\left(\hat{x}_a^* - \frac{\hat{x}_a + \hat{x}_b}{2}\right) + u(x_b, x_a^*) \\ &= \left(\frac{1}{2} + \frac{\kappa}{2} \frac{\hat{x}_a + \hat{x}_b}{\hat{x}_b - \hat{x}_a}\right)2(\hat{x}_a - \hat{x}_b)\left(\hat{x}_a^* - \frac{\hat{x}_a + \hat{x}_b}{2}\right) + u(x_b, x_a^*) \\ &= ((1 - \kappa)\hat{x}_a - (1 + \kappa)\hat{x}_b)\left(\hat{x}_a^* - \frac{\hat{x}_a + \hat{x}_b}{2}\right) + u(x_b, x_a^*). \end{aligned} \quad (32)$$

If x_a is a best response, then x_a is a local maximum of the pseudo utility $\tilde{U}_a(x_a, x_b)$. Then, the projection $\hat{x}_a = \text{proj}(x_a)$ must satisfy the first-order condition of (32). By Lemma 11 the unique $\hat{x}_a$ that satisfies the first-order condition is $\hat{x}_a = \hat{x}_a^* + \frac{\kappa}{1 - \kappa}\hat{x}_b$. Inserting $\hat{x}_a$ into the pseudo utility, we obtain

$$\begin{aligned} U_a(x_a, x_b) &= \left((1 - \kappa)\left(\hat{x}_a^* + \frac{\kappa}{1 - \kappa}\hat{x}_b\right) - (1 + \kappa)\hat{x}_b\right)\left(\hat{x}_a^* - \frac{\hat{x}_a^* + \frac{\kappa}{1 - \kappa}\hat{x}_b + \hat{x}_b}{2}\right) + u(x_b, x_a^*) \\ &= \left((1 - \kappa)\hat{x}_a^* - \hat{x}_b\right)\frac{1}{2}\left(\hat{x}_a^* - \frac{\hat{x}_b}{1 - \kappa}\right) + u(x_b, x_a^*) \\ &= \frac{1}{2(1 - \kappa)}(\hat{x}_b - (1 - \kappa)\hat{x}_a^*)^2 + u(x_b, x_a^*) \\ &= \frac{1}{2(1 - \kappa)}\frac{(x_b - x_a)^\top (x_b - (1 - \kappa)x_a^*)^2}{\|x_b - x_a\|} + u(x_b, x_a^*). \end{aligned} \quad (33)$$

If x_a is a local maximum of the pseudo utility, then the direction $\frac{x_a - x_b}{\|x_a - x_b\|}$ of x_a from x_b must be a local optimum of (33), given that the projection $\hat{x}_a$ on this direction is optimally chosen. The only local optimum is when $x_b - x_a$ is parallel to $x_b - (1 - \kappa)x_a^*$, that is, when x_a is on the line through $(1 - \kappa)x_a^*$ and x_b .

Together, we have shown that x_a is on the line through $(1 - \kappa)x_a^*$ and x_b and the scalar projection of x_a on this line is $\hat{x}_a^* + \frac{\kappa}{1 - \kappa}\hat{x}_b$. The unique point that satisfies these two conditions is $\tilde{x}_a$.

Sufficiency: By (33), if feasibility holds at $(\tilde{x}_a, x_b)$, then

$$U_a(\tilde{x}_a, x_b) = \frac{1}{2(1 - \kappa)}\|x_b - (1 - \kappa)x_a^*\|^2 + u(x_b, x_a^*).$$

For any $d \in \mathbb{R}^n$ with $\|d\| = 1$, define

$$B(d) := \frac{1}{2(1-\kappa)} (d^\top (x_b - (1-\kappa)x_a^*))^2 + u(x_b, x_a^*),$$

which is bounded above by $U_a(\tilde{x}_a, x_b)$ by the Cauchy-Schwarz inequality. The following lemma concludes the proof.

Lemma 14. *For any $x_a \in \mathbb{R}^n$, the utility $U_a(x_a, x_b)$ is bounded above by $B\left(\frac{x_b - x_a}{\|x_b - x_a\|}\right)$.*

Proof. Write $d(x_a) = \frac{x_b - x_a}{\|x_b - x_a\|}$. By (33), $B(d(x_a))$ is the pseudo utility $\tilde{U}_a(x_a, x_b)$ maximized over the scalar projection $\hat{x}_a$ in direction $d(x_a)$, so $\tilde{U}_a(x_a, x_b) \leq B(d(x_a))$.

If $u(x_a, x_a^*) \leq u(x_b, x_a^*)$, then $U_a(x_a, x_b) \leq u(x_b, x_a^*) \leq B(d(x_a))$. Assume for the rest of the proof that $u(x_a, x_a^*) > u(x_b, x_a^*)$.

If feasibility holds at (x_a, x_b) , then the vote share equals the pseudo vote share, so $U_a(x_a, x_b)$ equals the pseudo utility, which is bounded from above by $B(d(x_a))$.

If feasibility does not hold at (x_a, x_b) , then by Proposition 2 either the origin does not lie between the posterior means $\theta_a(x_a, x_b)$ and $\theta_b(x_a, x_b)$, or voters acquire a threshold signal. If the origin does not lie between $\theta_a(x_a, x_b)$ and $\theta_b(x_a, x_b)$, then one party receives all votes. If party a receives all votes, then $P_a(x_a, x_b) = 1 \leq \tilde{P}_a(x_a, x_b)$, so $U_a(x_a, x_b)$ is bounded above by the pseudo utility. If party b receives all votes, then $U_a(x_a, x_b) = u(x_b, x_a^*) \leq B(d(x_a))$. Finally, suppose voters acquire a threshold signal, so the origin lies between $\theta_a(x_a, x_b)$ and $\theta_b(x_a, x_b)$. We claim $\|x_a\| > \|x_b\|$. Suppose instead $\|x_a\| \leq \|x_b\|$. By the Cauchy-Schwarz inequality the scalar projections $\hat{x}_a$ and $\hat{x}_b$ onto the line through x_a and x_b satisfy $|\hat{x}_b| \leq \|x_b\|$ and $|\hat{x}_a| \leq \|x_a\| \leq \|x_b\|$, so $[\hat{x}_a, \hat{x}_b] \subseteq [-\|x_b\|, \|x_b\|]$. Because the density of μ is spherical, its projection onto every line through the origin is the same one-dimensional law, in which feasibility holds at $(-\|x_b\|, \|x_b\|)$ by feasibility at $(-x_b, x_b)$. Lemma 6 then gives feasibility at (x_a, x_b) , contradicting that feasibility does not hold there. Hence $\|x_a\| > \|x_b\|$, so $|\hat{x}_a| > |\hat{x}_b|$, and by Corollary 3(3), $P_a(x_a, x_b) < \tilde{P}_a(x_a, x_b)$. Together with $u(x_a, x_a^*) > u(x_b, x_a^*)$, $U_a(x_a, x_b)$ is bounded from above by the pseudo utility, which is bounded from above by $B(d(x_a))$. $\square$

This concludes the proof of sufficiency. $\square$

B.2.3 Equilibrium Verification

We use the best response lemma to verify that the polarizing platform pairs of Theorems 2 and 3 constitute equilibria.

Consider the symmetric setup with $\kappa < 1$, $\|x_a^*\| = \|x_b^*\|$, and $\frac{1-\kappa}{\kappa}\|x_a^*\| \leq \mathbb{E}[\|\theta_1\|]$. Set $x_b = (1-\kappa)x_b^*$. By $\|x_a^*\| = \|x_b^*\|$,

$$\text{Proj}_{(1-\kappa)x_a^*, x_b} \left(x_a^* + \frac{\kappa}{1-\kappa} x_b \right) = \text{Proj}_{(1-\kappa)x_a^*, (1-\kappa)x_b^*} (x_a^* + \kappa x_b^*) = (1-\kappa)x_a^*,$$

so the point $\tilde{x}_a$ of Lemma 13 equals $(1 - \kappa)x_a^*$. By $\frac{1-\kappa}{\kappa}\|x_a^*\| = \frac{1-\kappa}{\kappa}\|x_b^*\| \leq \mathbb{E}[\|\theta_1\|]$ and μ being spherical, the posterior means θ_a, θ_b under $(\tilde{x}_a, x_b)$ and under $(-x_b, x_b)$ are less far apart than those induced by the threshold signal informing voters whether $(x_b - x_a)^\top \theta \gtrless 0$. Because the induced posterior means are also symmetric around 0, feasibility holds at $(\tilde{x}_a, x_b)$ and $(-x_b, x_b)$. By the sufficiency part of Lemma 13, $x_a = (1 - \kappa)x_a^*$ is a best response to $x_b = (1 - \kappa)x_b^*$. By symmetry, x_b is a best response to x_a , so $(1 - \kappa)(x_a^*, x_b^*)$ is an equilibrium.

For the asymmetric setup, the positions $(x_a(\kappa), x_b(\kappa))$ satisfy

$$x_j = \text{Proj}_{(1-\kappa)x_j^*, x_{-j}} \left(x_j^* + \frac{\kappa}{1-\kappa} x_{-j} \right)$$

for $j \in \{a, b\}$ by Proposition 4. If $\max\{\|x_a(\kappa)\|, \|x_b(\kappa)\|\} \leq \kappa \mathbb{E}[\|\theta_1\|]$, then feasibility holds at $(x_a(\kappa), x_b(\kappa))$, $(-x_b(\kappa), x_b(\kappa))$, and $(x_a(\kappa), -x_a(\kappa))$ by Lemma 3 and Lemma 4, analogous to the argument in the equilibrium verification for Proposition 4. By the sufficiency part of Lemma 13 applied to each party, $(x_a(\kappa), x_b(\kappa))$ is an equilibrium.

Finally, we verify the formulas for the revealed ideal points and the vote share.

Derivation of θ_a, θ_b, P_a : Write $d = x_b^* - x_a^*$ and let $\Pi := \text{Proj}_{(1-\kappa)x_a^*, (1-\kappa)x_b^*}$ denote the orthogonal projection onto the line L through $(1 - \kappa)x_a^*$ and $(1 - \kappa)x_b^*$, which has direction d . The map Π is affine, so the difference of the party positions in (7) equals the projection onto L of the difference of the arguments, and that difference is $\frac{1-\kappa}{1-2\kappa}(1 - 2\kappa)d = (1 - \kappa)d$, which lies along L . Hence

$$x_b(\kappa) - x_a(\kappa) = (1 - \kappa)(x_b^* - x_a^*). \quad (34)$$

Because Π is affine and the two arguments average to $\frac{1-\kappa}{1-2\kappa} \frac{x_a^* + x_b^*}{2}$,

$$\frac{x_a(\kappa) + x_b(\kappa)}{2} = \Pi \left(\frac{1-\kappa}{1-2\kappa} \frac{x_a^* + x_b^*}{2} \right). \quad (35)$$

Orthogonal projection onto a line leaves the component along that line unchanged, so $d^\top \text{Proj}_{y, y+d}(w) = d^\top w$ for every y and w . Since $\text{Proj}_{0,d}(u) = \frac{d^\top u}{\|d\|^2} d$, this yields the parallel-projection identity

$$\text{Proj}_{0,d}(\text{Proj}_{y, y+d}(w)) = \text{Proj}_{0,d}(w). \quad (36)$$

By (34), $x_b(\kappa) - x_a(\kappa)$ is a positive multiple of d , so $\text{Proj}_{0, x_b(\kappa) - x_a(\kappa)} = \text{Proj}_{0,d}$ and $\frac{x_b(\kappa) - x_a(\kappa)}{2\kappa} = \frac{1-\kappa}{2\kappa} d$. Applying (36) to (35) with $y = (1 - \kappa)x_a^*$ and $w = \frac{1-\kappa}{1-2\kappa} \frac{x_a^* + x_b^*}{2}$,

$$\text{Proj}_{0, x_b(\kappa) - x_a(\kappa)} \left(\frac{x_a(\kappa) + x_b(\kappa)}{2} \right) = \text{Proj}_{0, x_b^* - x_a^*} \left(\frac{1-\kappa}{1-2\kappa} \frac{x_a^* + x_b^*}{2} \right). \quad (37)$$

Substituting (37) and $\frac{x_b(\kappa) - x_a(\kappa)}{2\kappa} = \frac{1-\kappa}{2\kappa} (x_b^* - x_a^*)$ into (6) gives, for $j \in \{a, b\}$,

$$\theta_j(\kappa) = \text{Proj}_{0, x_b^* - x_a^*} \left(\frac{1-\kappa}{1-2\kappa} \frac{x_a^* + x_b^*}{2} \right) + \frac{1-\kappa}{2\kappa} (x_j^* - x_{-j}^*). \quad (38)$$

For the vote share, $\|x_b\|^2 - \|x_a\|^2 = (x_b - x_a)^\top (x_a + x_b)$. Using (34), then $d^\top \text{Proj}_{y, y+d}(w) = d^\top w$

on (35),

$$(x_b^* - x_a^*)^\top \frac{x_a(\kappa) + x_b(\kappa)}{2} = \frac{1-\kappa}{1-2\kappa} (x_b^* - x_a^*)^\top \frac{x_a^* + x_b^*}{2} = \frac{1-\kappa}{1-2\kappa} \frac{\|x_b^*\|^2 - \|x_a^*\|^2}{2}, \quad (39)$$

so that

$$\|x_b(\kappa)\|^2 - \|x_a(\kappa)\|^2 = \frac{(1-\kappa)^2}{1-2\kappa} (\|x_b^*\|^2 - \|x_a^*\|^2). \quad (40)$$

With $\|x_b(\kappa) - x_a(\kappa)\|^2 = (1-\kappa)^2 \|x_b^* - x_a^*\|^2$ from (34), the vote-share expression $P_a = \frac{1}{2} + \frac{\kappa}{2} \frac{\|x_b\|^2 - \|x_a\|^2}{\|x_b - x_a\|^2}$ becomes

$$P_a(\kappa) = \frac{1}{2} + \frac{\kappa}{2(1-2\kappa)} \frac{\|x_b^*\|^2 - \|x_a^*\|^2}{\|x_b^* - x_a^*\|^2}. \quad (41)$$

B.2.4 Uniqueness

Let $\kappa < 1$. We prove uniqueness in two steps, under $\|x_b^* - x_a^*\| < \kappa \mathbb{E}[\|\theta_1\|]$. First, we show feasibility is slack in an equilibrium. Second, we apply the necessity part of the multidimensional best response lemma to reduce the game to one dimension and invoke the one-dimensional uniqueness results. Let (x_a, x_b) be an equilibrium.

First step: feasibility is slack. Denote scalar projections onto the direction $x_b - x_a$ by a hat, such as $\hat{x}_a = \text{proj}_{x_b - x_a}(x_a)$ and $\hat{x}_a^* = \text{proj}_{x_b - x_a}(x_a^*)$. Restricting both parties to positions on the line through x_a and x_b leaves (x_a, x_b) an equilibrium, so by Lemma 12 the pair $(\hat{x}_a, \hat{x}_b)$ is an equilibrium of the induced one-dimensional game with voter prior $\hat{\mu}$. Because μ is spherical, its scalar projection onto every line through the origin is the same one-dimensional distribution $\hat{\mu}$, with

$$\Delta := \mathbb{E}_{\hat{\theta} \sim \hat{\mu}}[\hat{\theta} \mid \hat{\theta} > 0] - \mathbb{E}_{\hat{\theta} \sim \hat{\mu}}[\hat{\theta} \mid \hat{\theta} < 0] = 2\mathbb{E}[\|\theta_1\|].$$

By Lemmas 8 and 10, $|\hat{x}_b - \hat{x}_a| \leq 2|\hat{x}_b^* - \hat{x}_a^*| \leq 2\|x_b^* - x_a^*\|$. Together with $\|x_b^* - x_a^*\| < \kappa \mathbb{E}[\|\theta_1\|]$,

$$|\hat{x}_b - \hat{x}_a| < 2\kappa \mathbb{E}[\|\theta_1\|] = \kappa \Delta.$$

If $0 \in (\theta_a(\hat{x}_a, \hat{x}_b), \theta_b(\hat{x}_a, \hat{x}_b))$, then by Lemma 5 feasibility is slack at $(\hat{x}_a, \hat{x}_b)$ and therefore at (x_a, x_b) . Otherwise, if $x_a \neq x_b$, one party receives all votes by Corollary 3, which violates our equilibrium definition. We now rule out $x_a = x_b$.

First, $(x_a, x_b) = (0, 0)$ is not an equilibrium for $\kappa < 1$. By $x_a^* \neq x_b^*$, at least one ideal policy is nonzero, say $x_a^* \neq 0$. Moving toward x_a^* gives party a a positive vote share for $\kappa < 1$ and a policy closer to x_a^* when winning, a profitable deviation.

Second, we rule out $(x_a, x_b) = (x, x)$ with $x \neq 0$.

Lemma 15. *For any κ , if $x \neq 0$ does not lie on both the line through 0 and x_a^* and the line through 0 and x_b^* , then (x, x) is not an equilibrium.*

Proof. Let $x_a = x_b = x \neq 0$. If x does not lie on the line through 0 and x_a^* , then every ε -neighborhood of x contains a point x' closer to both 0 and x_a^* , obtained by moving x in the direction $\frac{1}{2}(x_a^* - x) + \frac{1}{2}(0 - x)$. For small $\varepsilon > 0$, adopting x' gives party a all votes and a higher

policy utility when winning, a profitable deviation. The argument for party b and the line through 0 and x_b^* is analogous. $\square$

If 0, x_a^* , and x_b^* are not jointly on a line, the two lines meet only at 0, so every $x \neq 0$ satisfies the hypothesis of Lemma 15 and (x, x) is not an equilibrium. If instead 0, x_a^* , and x_b^* are jointly on a line and $x \neq 0$ lies on this line, then 0 lies strictly between the projections of x_a^* and x_b^* : for Theorem 2 by $\|x_a^*\| = \|x_b^*\|$ and $x_a^* \neq x_b^*$, and for Theorem 3 by assumption. As in the proof of Lemma 15, at least one party then profits from moving slightly toward the origin, obtaining all votes and a policy closer to its ideal point. Hence $x_a = x_b$ does not occur in equilibrium, and feasibility is slack.

Second step: reduction to one dimension. By the first step, feasibility is slack in any equilibrium (x_a, x_b) . The necessity part of the multidimensional best response lemma then places x_a and x_b on the line through $(1 - \kappa)x_a^*$ and $(1 - \kappa)x_b^*$. On that line the induced game is the one-dimensional game by Lemma 12, so the one-dimensional uniqueness results in Propositions 3 and 4 apply.

B.2.5 Equilibrium if $\kappa \geq 1$

Define $\tilde{x}_a^* := \text{proj}_{x_b^* - x_a^*}(x_a^*)$ and $\tilde{x}_b^* := \text{proj}_{x_b^* - x_a^*}(x_b^*)$. The ordering $\tilde{x}_a^* < 0 < \tilde{x}_b^*$ is assumed in Theorem 3 and implied by $\|x_a^*\| = \|x_b^*\|$ and $x_a^* \neq x_b^*$ in Theorem 2. We establish if $\kappa \geq 1$ and $\tilde{x}_a^* < 0 < \tilde{x}_b^*$, the unique equilibrium is $(0, 0)$.

First, $(0, 0)$ is an equilibrium. The reason is that under $\kappa \geq 1$, the implemented policy is 0 if one party chooses position 0. This follows from Corollary 3 and the fact that $\theta_a(x_a, x_b)$ and $\theta_b(x_a, x_b)$ are weakly between x_a and x_b for $\kappa \geq 1$. Hence, no party has a profitable deviation.

Second, we show uniqueness. Suppose, $(x_a, x_b) \neq (0, 0)$ is an equilibrium.

Consider first $x_a = x_b = x \neq 0$. If the origin, x_a^* , and x_b^* are not on a joint line, then x fails to lie on at least one of the lines through the origin and x_a^* and through the origin and x_b^* , so (x, x) is not an equilibrium by Lemma 15. If instead the origin, x_a^* , and x_b^* lie on a joint line and x lies on this line, then $\tilde{x}_a^* < 0 < \tilde{x}_b^*$. The party whose ideal point lies on the opposite side of the origin from x then profits from moving slightly toward the origin, obtaining all votes and a policy closer to its ideal point. So $x_a = x_b \neq 0$ does not occur in equilibrium.

Consider now $x_a \neq x_b$ and define $\hat{x}_a^* := \text{proj}_{x_b - x_a}(x_a^*)$ and $\hat{x}_b^* := \text{proj}_{x_b - x_a}(x_b^*)$. By Lemma 12, $(\hat{x}_a, \hat{x}_b) = (\text{proj}_{x_b - x_a}(x_a), \text{proj}_{x_b - x_a}(x_b))$ is an equilibrium of the induced one-dimensional game with ideal points $\hat{x}_a^*, \hat{x}_b^*$. If $\hat{x}_a^* \leq 0 \leq \hat{x}_b^*$, then $(x_a, x_b) = (0, 0)$ by Lemma 10. Otherwise, the origin does not lie between $\hat{x}_a^*$ and $\hat{x}_b^*$. Without loss, suppose, $0 < \hat{x}_a^*$ and $0 < \hat{x}_b^*$. By Lemma 8 (3) and $x_a \neq x_b$,

$$\hat{x}_a^* \leq \hat{x}_a < \hat{x}_b \leq 2\hat{x}_b^* - \hat{x}_a^*.$$

Thus $0 < \hat{x}_a < \hat{x}_b$, so for $\kappa \geq 1$ the posterior means θ_a, θ_b lie weakly between $\hat{x}_a$ and $\hat{x}_b$, so party a obtains all the votes, contradicting our equilibrium definition.

B.2.6 Corollary 2

Proof. Denote the equilibrium positions by $x_j(\kappa) := (1 - \kappa)x_j^*$ for $j \in \{a, b\}$. For any $\kappa \in (\underline{\kappa}, 1]$, $P_a(x_a(\kappa), x_b(\kappa)) = 1/2$ and by μ being spherical $\mathbb{E}[u(x_a(\kappa), \theta)] = \mathbb{E}[u(x_b(\kappa), \theta)]$. By a bias-variance decomposition, aggregate policy utility is

$$-\mathbb{E}[\|\theta - (1 - \kappa)x_a^*\|^2] = -\text{Var}[\theta] - (1 - \kappa)^2\|x_a^*\|^2,$$

which is increasing in κ and strictly increasing in κ if $x_a^* \neq 0$. By $\|x_a^*\| = \|x_b^*\|$ and $x_a^* \neq x_b^*$, we have $x_a^* \neq 0$.

The total information cost is

$$\begin{aligned} \kappa c(\rho) &= \kappa \mathbb{E}_{\theta \sim \rho} [\|\theta\|^2] = \kappa \left\| \text{Proj}_{0, x_b^* - x_a^*} \left(\frac{x_a(\kappa)}{\kappa} \right) \right\|^2 \\ &= \kappa \left\| \text{Proj}_{0, x_b^* - x_a^*} \left(\frac{(1 - \kappa)x_a^*}{\kappa} \right) \right\|^2 = \frac{(1 - \kappa)^2}{\kappa} \left\| \text{Proj}_{0, x_b^* - x_a^*} (x_a^*) \right\|^2. \end{aligned}$$

Thus the information cost is decreasing in κ for $\kappa < 1$. By $\|x_a^*\| = \|x_b^*\|$ and $x_a^* \neq x_b^*$, $\text{Proj}_{0, x_b^* - x_a^*} (x_a^*) \neq 0$, so the information cost is strictly decreasing in κ . Aggregate policy utility strictly increases in κ and the information cost strictly decreases in κ , so aggregate voter welfare strictly increases in κ on $(\underline{\kappa}, 1]$. $\square$

C Online Appendix: Other Results and Proofs

C.1 Imperfect Observation of Party Platforms

Our results on voter learning are, under some assumptions, robust to voters observing a *noisy* signal about party platforms before learning about ideal points (under a timing where parties move before voters learn).

Suppose that platforms are stochastic and independent of each other and of voters' ideal points. This stochasticity may stem from parties making random errors when choosing their platforms or from parties having stochastic and private ideal points, similar to Matějka and Tabellini (2021). Further, assume that, after a common signal about platforms, voters acquire a signal about their ideal points. Because both signals preserve the independence of platforms and ideal points, the expected policy utility from voting for unknown platform x under unknown ideal point θ can be written as

$$\mathbb{E}[u(x, \theta)] = u(\mathbb{E}[x], \mathbb{E}[\theta]) - \mathbb{E}[(x - \mathbb{E}[x])^\top A(x - \mathbb{E}[x])] - \mathbb{E}[(\theta - \mathbb{E}[\theta])^\top A(\theta - \mathbb{E}[\theta])].$$

Up to the constant $\mathbb{E}[(x - \mathbb{E}[x])^\top A(x - \mathbb{E}[x])]$, the agent's utility is as under known platforms, (3), except for replacing the platform x with the expected platform $\mathbb{E}[x]$. Thus, our results from section 3 remain to hold after replacing party platforms with their expectation.

If voters observe heterogeneous signals about platforms, this creates heterogeneity in their learning strategies, such as the direction in which they learn (cf. Theorem 1). However, as long as their signals about platforms are similar enough, our results should carry over approximately. As a consequence, an extension of our model to heterogeneous signal may explain the empirical finding that the ideal points of politically better informed citizens are better described by a low-dimensional model (Converse 1964; Hare 2022), since better informed voters should have more homogeneous information about platforms.

C.2 Corollary 1

Proof. Define

$$V := \text{span}(\{A(x_j - x_1)\}_{j=2, \dots, k}),$$

the subspace of dimension at most $k - 1$ through the prior mean from the statement of the corollary. As in the proof of Theorem 1, write

$$c(\rho) = \mathbb{E}_{\theta \sim \rho}[\theta^\top \theta]$$

for the cost of $\rho \in \Delta(\mathbb{R}^n)$, and say ρ is feasible if $\rho \preceq_{\text{MPS}} \mu$.

We first show the instrumental value of information depends only on the projection of the

posterior mean onto V . Analogously to the two-party objective (8),

$$\begin{aligned}
& \mathbb{E}_\tau \left[\max_j \mathbb{E}_\pi \left[-(\theta - x_j)^\top A(\theta - x_j) \right] \right] \\
&= \mathbb{E}_\tau \left[\max_j \left\{ -x_j^\top A x_j + 2x_j^\top A \mathbb{E}_\pi[\theta] \right\} \right] - \mathbb{E}_\mu[\theta^\top A \theta] \\
&= \mathbb{E}_\tau \left[\max_j \left\{ -x_j^\top A x_j + 2(x_j - x_1)^\top A \mathbb{E}_\pi[\theta] \right\} \right] + C,
\end{aligned}$$

where we use the law of iterated expectations and the second inequality adds and subtracts $2x_1^\top A \mathbb{E}_\pi[\theta]$ so that

$$C = 2x_1^\top A \mathbb{E}_\mu[\theta] - \mathbb{E}_\mu[\theta^\top A \theta]$$

does not depend on τ and can be ignored. Each term $(x_j - x_1)^\top A \mathbb{E}_\pi[\theta]$ depends on $\mathbb{E}_\pi[\theta]$ only through its projection onto V , so only this projection is payoff-relevant.

The four steps of the proof of Theorem 1 now apply with the following modifications. Step 1 reflects ρ across V instead of L : ref is replaced by the orthogonal reflection across V ,

$$\text{ref}_k(\theta) := 2 \text{Proj}_V(\theta) - \theta,$$

where Proj_V denotes the orthogonal projection onto V . The reflection of ρ is feasible because ref_k is orthogonal and μ is spherical, it preserves the instrumental value because it fixes the projection onto V , and it preserves the cost because it preserves $\theta^\top \theta$. Step 2 mixes ρ with $\text{ref}_k(\rho)$, yielding a feasible distribution symmetric about V with the same value and cost. Step 3 projects this mixture orthogonally onto V , which by the symmetry of the mixture is a mean-preserving contraction, preserving the instrumental value while weakly reducing the cost. Step 4 garbles according to the action recommendation, partitioning the state space into the k cells on which each platform is optimal and replacing the distribution by its conditional means on these cells. This garbling preserves the instrumental value, weakly reduces the cost, and produces a distribution supported on at most k points in V . As in the proof of Theorem 1, Step 3 or 4 strictly reduce the information cost unless ρ is supported on at most k points, contained in V .

Hence any solution is supported on at most k revealed ideal points, contained in V . $\square$

C.3 Proposition 5

Upon closer examination, it is not clear whether the prediction of Theorem 1 aligns with the evidence that survey responses are well explained by a one-dimensional spatial model in section 1.1.1. First, would it not be sufficient for voter ideal points to be on some one-dimensional *curve* for a one-dimensional spatial model to explain voter's survey responses? In that case, Theorem 1 would be proving too much. Second, does equivalence to a one-dimensional spatial model require not just that the ideal points but also the *policies* are in a one-dimensional space? Theorem 1 predicts one-dimensional ideal points within a multidimensional policy space, while in one-dimensional spatial models *both* the ideal points as well as the policies live in a one-dimensional space. If the answer to the second question is affirmative, then Theorem 1 would be proving too little to explain the

evidence.

In the following, we show neither is the case and Theorem 1 proves the property of voter ideology identified by the evidence, namely the property that ensures that voters' survey responses can be explained by a one-dimensional spatial model.

First, we need additional definitions. A *multidimensional spatial model* with quadratic utility is defined identically to the one-dimensional spatial model described above, except for the parameters $\{\theta_i, x_{j1}, x_{j2}\}$ being elements of $\mathbb{R}^n$ and $u(x, \theta) = -(x - \theta)^\top (x - \theta)$ being the multidimensional analogue of quadratic utility.¹¹ It turns out that the property of a multidimensional spatial model identified by the evidence is that respondents' *ideal points are on a line when projected onto the space spanned by the survey questions*. Formally, this property states that there exist $\lambda_i \in \mathbb{R}$, $\Delta\theta \in \mathbb{R}^n$ and $\theta_i^\perp \in \mathbb{R}^n$, such that

$$\begin{aligned}\forall i: \theta_i &= \theta_1 + \lambda_i \Delta\theta + \theta_i^\perp, \\ \forall i, j: (x_{j1} - x_{j2})^\top \theta_i^\perp &= 0.\end{aligned}$$

That is, each ideal point θ_i is on the line through θ_1 with direction $\Delta\theta$, modulo a component $\theta_i^\perp$ that is orthogonal to the policy-differences $x_{j1} - x_{j2}$ for each question j . We say that a multidimensional spatial model with (n -dimensional) parameters $\{\theta_i, x_{j1}, x_{j2}\}$ is *observationally equivalent* to a one-dimensional spatial with (one-dimensional) parameters $\{\hat{\theta}_i, \hat{x}_{j1}, \hat{x}_{j2}\}$ if they predict the same likelihood $\Pr(y_{ij} = 1)$ over survey responses for all i and j .

Proposition 5. *Under quadratic utility, a multidimensional spatial model is observationally equivalent to some one-dimensional spatial model if and only if the multidimensional ideal points are on a line when projected onto the space spanned by the survey questions.*

Proposition 5 shows ideal points being on a line (when projected onto the space spanned by the survey questions) is the property of ideal points that makes survey behavior explainable by a one-dimensional spatial model. The parenthesized caveat holds because, naturally, survey responses are not affected by policy dimensions that are orthogonal to all survey questions. Because voter surveys try to cover most relevant policy dimensions, we take this caveat to be of limited importance. Since the above-mentioned papers show voters' survey responses are well-explained by a one-dimensional spatial model (and not much better by higher-dimensional models), we conclude they confirm the prediction of Theorem 1.

The high-level intuition for Proposition 5 is as follows. For the “only if”-direction, suppose voter ideal points were not on a line but, say, on a U-shaped curve. Then, the extreme voters on both sides of the U may prefer some policy to a policy at the bottom of the U, that is preferred by the centrist voters. Such non-monotonic behavior is ruled out by one-dimensional ideological spectrum. For the “if”-direction, for any survey question, one can find suitable projections of its two policies onto the voter line that do not change predicted behavior and make the model one-dimensional.

¹¹The result remains the same if we assume a general quadratic form $u(x, \theta) = (x - \theta)^\top A(x - \theta)$ because we can switch to an orthonormal basis of A .

For brevity, we refer by property L to the property that ideal points are on a line when projected onto the space spanned by the survey questions, as defined in the main text.

It is known (e.g., Ladha 1991) that the one-dimensional spatial model with quadratic utility is equivalent to the one-dimensional item-response theory (IRT) model, which the empirical papers we referred to estimate (Jessee 2009; Jessee 2012; Tausanovitch and Warshaw 2012; Shor and Rogowski 2018; Fowler, Hill, Lewis, Tausanovitch, Vavreck, and Warshaw 2022).

The IRT model is given as follows. Let y_{ij} denote the response of individual i to question j , which can be either 1 or 0. Under a one-dimensional IRT model, the likelihood is given by

$$\Pr(y_{ij} = 1) = \Phi(\alpha_j + \beta_j p_i),$$

where Φ is the logistic or the normal cumulative distribution function, $\alpha_j, \beta_j \in \mathbb{R}$ are question-specific parameters, and $p_i \in \mathbb{R}$ are individual-specific parameters.

In our proof, we show (Part I) if a multidimensional spatial model has property L , then there is a one-dimensional IRT model that is observationally equivalent, and (Part II) if there is a one-dimensional IRT model that is observationally equivalent to a given multidimensional spatial model, then the spatial model satisfies property L . Because of the equivalent to of one-dimensional IRT models to one-dimensional spatial models with quadratic utility, this establishes our result.

Proof. First, in the spatial model, we have

$$\|x_{j1} - \theta_i\|^2 - \|x_{j2} - \theta_i\|^2 = (\Delta x_j)^\top (\theta_i - \bar{x}_j)$$

where $\Delta x_j := x_{j1} - x_{j2}$ and $\bar{x}_j := \frac{x_{j1} + x_{j2}}{2}$. We suppose that the same response item is not part of two different questions, which seems to be satisfied in practice, so there are no restrictions that x_{j1} or x_{j2} for different j are the same.¹² Hence, there are no restrictions on Δx_j and $\bar{x}_j$, and we can reparametrize the model through $\{\Delta x_j, \bar{x}_j, \theta_i\}$ instead of $\{x_{j1}, x_{j2}, \theta_i\}$.

The multidimensional spatial model with parameters $\{\Delta x_j, \bar{x}_j, \theta_i\}$ is observationally equivalent to the one-dimensional IRT model with parameters $\{\alpha_j, \beta_j, p_i\}$ if and only if

$$\forall i, j: (\Delta x_j)^\top (\theta_i - \bar{x}_j) = \alpha_j + \beta_j p_i \quad (42)$$

Part I Suppose the multidimensional spatial model satisfies property L . Then, define

$$\begin{aligned} p_i &:= \lambda_i \\ \beta_j &:= (\Delta x_j)^\top \Delta \theta \\ \alpha_j &:= (\Delta x_j)^\top (\theta_1 - \bar{x}_j) \end{aligned}$$

Then, (42) holds by

$$(\Delta x_j)^\top (\theta_j - \bar{x}_j) = (\Delta x_j)^\top (\theta_1 - \bar{x}_j + \lambda_i \Delta \theta + \theta_i^\perp) = \alpha_j + \beta_j \lambda_i$$

¹²This condition is sufficient but not necessary for our proof.

and the IRT model with parameters $\{\alpha_j, \beta_j, p_i\}$ is observationally equivalent.

Part II If there is a one-dimensional model that is observationally equivalent to the multidimensional spatial model, then (42) holds, which implies

$$(\Delta x_j)^\top (\theta_i - \theta_1) = \beta_j (p_i - p_1).$$

Take $i = 2$, then the projection of $\theta_2 - \theta_1$ on all Δx_j is given, so $\theta_2 - \theta_1$ is uniquely pinned down in the space spanned by $\{\Delta x_j\}$. For any other $i > 2$, $\theta_i - \theta_1$ is $\beta_j (p_i - p_1) = \frac{p_i - p_1}{p_2 - p_1} (\Delta x_j)^\top (\theta_2 - \theta_1)$. Thus, the projection of $\theta_i - \theta_1$ on the space spanned by $\{\Delta x_j\}$ is a multiple of the one of $\theta_2 - \theta_1$. Thus, property L holds. $\square$